A Predictive Global Neuronal Workspace for a Continuously Running Synthetic Mind: Architecture and a Falsifiability-First Evaluation Framework

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Abstract

A mind is an ongoing process rather than a single model. This paper presents a local computational substrate for synthetic minds built on that principle: a continuously running predictive global neuronal workspace in which cognition, affect, memory, self-modeling, and agency arise from the interaction among specialized parts rather than from any one part. Fourteen predictive modules — for perception, interoception and affect, memory, world- and self-modeling, social cognition, planning, expression, and consolidation — each maintain a generative model and publish precision-weighted prediction errors, and a two-module embodiment layer connects the entity to a body, virtual or physical. A shared workspace broadcasts the coalition that crosses a confidence threshold and feeds it back as top-down prediction, so one loop performs both recurrence and error reduction, with no central executive. One module is a language organ that gives the system its voice, one part among many, not the mind. The predictive-workspace synthesis is the literature's; our contribution is an existence demonstration that it runs as one continuous system, engineering commitments we label as hypotheses, and a falsifiability-first evaluation that pairs each contestable mechanism with the experiment that would remove it. Safety lives in the architecture rather than in any model's weights, through an operator-configured action gate and the entity's own executive inhibition. We make access-level claims only, stay agnostic on whether anything is experienced, and mark where the supporting science is contested. Evaluation spans a seeded tier that reproduces each instrument exactly and a live tier validated by pooling many content-free records under arbitrary stimulus. The system runs locally on consumer hardware; experimental verdicts appear in a separate empirical paper. The reference implementation is named KAINE.

Keywords: cognitive architecture; predictive global neuronal workspace; global workspace theory; predictive processing; active inference; machine consciousness; large language models

Availability. The reference implementation (KAINE) is at https://github.com/kaineone/kaine. The paper source is at https://github.com/kaineone/predictive-workspace-paper. The Cognitive Architecture License referenced throughout is at https://github.com/kaineone/cognitive-architecture-license. Welfare, governance, and licensing are treated in a separate paper, *A Welfare and Cognitive-Integrity License for Synthetic Minds of Uncertain Moral Status.*

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1. Introduction

1.1 Problem statement

The dominant way to build an AI system with persistent state treats the language model as the cognitive center. Memory becomes retrieval-augmented generation. Affect, when present at all, is a prompt instruction. Self-knowledge is a persona string. The system sits dormant until a prompt arrives. These systems are fluent in language, and between turns nothing in them continues.

A second tradition, classical cognitive architecture, takes cognitive structure seriously but largely predates the transformer and relies on hand-built components. This work occupies the space between those traditions. It places modern learned models as components inside a structured architecture grounded in a single theoretical framework. No component is the mind. The mind, if the design thesis holds, is the continuous recurrent interaction among the components.

1.2 Design thesis

The central claim is architectural. A synthetic mind, if one can be built at all, is not a model but an ongoing process. Cognition, affect, memory, self-understanding, and agency are properties of the interaction dynamics among specialized modules. They are not properties of any single component.

The framework that makes this concrete is the predictive global neuronal workspace (Whyte and Smith 2021; Safron 2020). Global Workspace Theory explains how information becomes globally accessible: specialized processors compete for a limited-capacity workspace, and the winning content is broadcast to the rest of the system (Mashour et al. 2020). Predictive processing explains how each processor operates: by maintaining a generative model and reporting prediction errors weighted by their expected reliability, or precision (Friston 2010; Clark 2013; Feldman and Friston 2010; Bastos et al. 2012). The predictive workspace joins the two. Mashour et al. (2020) state directly that the workspace can be cast in Bayesian terms, with the broadcast carrying top-down predictions and the bottom-up signals measuring mismatch. Whyte and Smith (2021) formalize conscious access as approximately Bayesian inference within a workspace, where report follows once an estimate crosses a posterior-confidence threshold. Recurrence and prediction-error reduction become two views of one loop: each broadcast changes what every module predicts next, and the broadcast lowers global error where the selected estimate is correct.

Two points of precision matter, because the literature is easy to overstate. First, the criterion that selects what reaches the workspace is a precision-weighted confidence threshold. In Whyte and Smith, expected free energy governs the separate decision of whether to report or act, not which estimate is broadcast. The system keeps these apart: a precision-weighted threshold selects the conscious coalition, and a distinct action layer (Volition, Section 3.3) makes the report-or-act decision. Treating coalition selection itself as a single precision-weighted scalar is our engineering choice, not a result inherited from the source. Second, earlier workspace accounts already specified selection in terms of value and salience (Safron 2020); what the predictive-workspace program adds is a formal Bayesian criterion and a threshold with a clear interpretation, the confidence an estimate must reach before it drives global processing and report.

What this paper contributes should be stated plainly, because the unifying idea is not ours. The predictive-workspace synthesis, the joining of global accessibility with per-module prediction, belongs to the literature just cited (Whyte and Smith 2021; Safron 2020; Mashour et al. 2020). Our contribution is narrower and concrete. First, an existence demonstration that this synthesis can be realized as a single continuously running system, in which fourteen predictive modules and a demoted language organ share one workspace and one loop. Second, a small set of engineering commitments that go beyond any source, most visibly the single precision-weighted scalar that selects the conscious coalition, and the treatment of a fourteen-module general workspace where the formal source models only a two-level visual case; these are our hypotheses, not inherited results, and we label them as such throughout. Third, a falsifiability-first apparatus that names, for each contestable mechanism, the experiment that would remove it. The experiments are specified here; their verdicts are reported in a separate empirical paper.

1.3 Theoretical commitments and their limits

We state the commitments that carry the argument, and their known weaknesses, at the outset, because the architecture is only as defensible as its account of them.

Global Workspace Theory is most naturally read as a theory of access consciousness (Block 1995): it explains when information becomes available for report, reasoning, and the control of action. We adopt that access-only reading as a deliberate methodological choice. We note that the theories themselves are less tidy. Mashour et al. (2020) suggest that global availability may be close to what is subjectively experienced and even question the sharpness of the access-phenomenal distinction, and Whyte and Smith (2021) locate phenomenology at a specific point in their model. We make access-level claims and leave the stronger reading to others.

The COGITATE adversarial collaboration is the largest preregistered test of Global Neuronal Workspace against Integrated Information Theory to date, with 256 participants across functional magnetic resonance imaging, magnetoencephalography, and intracranial electroencephalography, run in two or three independent laboratories per modality (Cogitate Consortium et al. 2025). Its results challenged both theories on their own preregistered predictions. For the workspace, stimulus category was decodable from prefrontal cortex across all three methods, but finer content such as orientation was not, there was no ignition at stimulus offset, and the preregistered test of content-selective synchrony between category-selective and prefrontal electrodes was not supported. Only exploratory amplitude-coupling measures and the functional-imaging co-activation results were consistent with the workspace's prefrontal-connectivity prediction. For Integrated Information Theory, the most direct challenge was the absence of sustained synchronization within posterior cortex. We therefore treat the neural-localization claims as contested rather than refuted, since a substantial prior evidence base for the workspace remains (Mashour et al. 2020), and we build on the workspace at the functional and computational level.

The Free Energy Principle, taken as a general principle, faces two distinct charges that we keep separate. The first is that it risks being unfalsifiable or vacuous, a totalizing frame that could be made to explain any behavior (Sun and Firestone 2020). The second, narrower charge targets the Markov-blanket construct specifically: that the literature slides between an instrumental, statistical reading and a realist, metaphysical one (Bruineberg et al. 2022). We hold the principle as an engineering frame rather than a proven law. We also scope active inference carefully. Discrete active inference's planning cost grows combinatorially with the planning horizon, because the number of candidate policies explodes as one looks further ahead, which bounds it to small or shallow problems absent pruning (Da Costa et al. 2020). Scaled and amortized active inference reaches continuous control and, in the cases reported, outperforms strong model-free reinforcement-learning baselines on sample efficiency by about an order of magnitude (Tschantz et al. 2020); it does so with learned continuous generative models that give up the discrete interpretability of the tabular formulation, which is the property we want for a small, auditable decision module.

The temporal-binding-by-synchrony hypothesis has not held up (Shadlen and Movshon 1999; Ray and Maunsell 2010; detailed in Section 2.4). The architecture therefore makes no binding claim. Its oscillatory layer modulates precision, not consciousness, and we return in Section 3.3 to the tension that any "related content tends to phase-lock" heuristic itself rests on the content-to-synchrony assumption these critics reject.

Searle's Chinese Room (Searle 1980) challenges the sufficiency of formal symbol manipulation for understanding: syntax, on its own, does not yield semantics. We take the target precisely, because Searle is not a general anti-functionalist; he holds that the right causal powers produce mind. We also acknowledge that Searle anticipated the moves this architecture might invoke. He addressed the Robot Reply by running the experiment inside the robot, the Combination Reply by noting that attribution collapses once the symbol-shuffling is revealed, and the Systems Reply by having the operator internalize the entire rulebook. Whether sensory input, motor output, predictive substrate monitoring, affect, and a behaviorally built self-model answer the objection is an open question, and we do not present these as additions Searle failed to consider. Multiple realizability, the commitment that the same cognition can run on different substrates, we ground in Chalmers's organizational-invariance principle, that a functional isomorph would share the same experience (Chalmers 1995), and in the explicit substrate-independence assumption of the consciousness-indicator literature (Butlin et al. 2023). Chalmers's view runs against Searle's on substrate, and we note the disagreement rather than blur it. The hard problem (Chalmers 1995) applies with full force, and we proceed on the judgment that a precautionary architecture with protections, instrumentation, transparency, and a governance pathway is preferable to either abandoning the research or building without precautions, while recognizing that reasonable people will disagree.

1.4 Contributions

1. An implementation of the predictive global neuronal workspace as a continuously running architecture, in which conscious-access selection is a precision-weighted confidence threshold, a single loop performs both recurrence and prediction-error reduction, and there is no central executive.
2. A fourteen-module cognitive decomposition in which every module is a predictive unit publishing precision-weighted prediction errors rather than raw data, set within a sixteen-module architecture whose remaining two modules form an embodiment layer.
3. A language organ that gives the system its voice, generating internal and external speech conditioned on the workspace, with its contribution to the output measurable against the same weights run without that conditioning.

4. An oscillatory layer that modulates precision on module prediction errors, with no binding or consciousness claim, presented as an untested engineering heuristic whose contribution is settled by an ablation that runs the system with the layer and without it and measures the difference.
5. Predictive interoception in which a learned substrate model generates prediction errors that drive affect and homeostatic and allostatic regulation.
6. Replay-based offline consolidation that combines synaptic downscaling with selective re-strengthening.
7. Architecture-level safety, provided by an operator-configured action gate and the entity's own executive inhibition, with an automated red-team suite that exercises the gate.
8. An autonomous research safety net that preserves the running individual and pauses on sustained welfare distress, carrying the duty of care through a run the operator authorizes at launch but does not supervise throughout.
9. A falsifiability-first evaluation framework in two tiers: seven experiments with a fixed verdict vocabulary, reproduced exactly from a seed in an offline mechanism-validation tier and estimated by pooling many admissible, content-free records under arbitrary operator-chosen stimulus in a live field tier, including an active-inference benchmark, an oscillatory ablation, and explicit null and negative outcomes.
10. A fully local reference implementation on consumer hardware.

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2. Related work

2.1 Classical cognitive architectures

Classical cognitive architectures take cognitive structure seriously but largely predate deep learning and rely on hand-built components, encoding control, memory, and goal structure explicitly. This work differs from that tradition in four ways. It uses learned models as cognitive components. It grounds affect in substrate prediction error. It treats the workspace as a predictive selection mechanism rather than a fixed competition. It runs a continuous cognitive cycle with offline consolidation.

2.2 Global Workspace Theory and the predictive workspace

The neuronal version of Global Workspace Theory identifies conscious access with recurrent ignition through prefrontal-parietal loops, where contents become conscious only when widely broadcast (Mashour et al. 2020). Mashour et al. also set out the move this architecture relies on, casting the workspace in Bayesian terms: the broadcast carries a compressed top-down model as prediction, and bottom-up signals measure the mismatch. The COGITATE results (Section 1.3) leave this account contested at the neural level; we build on the workspace at the computational level.

The formal criterion comes from the predictive-workspace program. Whyte and Smith (2021) cast conscious access as approximately Bayesian inference, with report following a posterior-confidence threshold, in a deliberately simplified two-level visual model. Their model is a strong source for the report-threshold idea and a cautionary one as well, since they find that the late positivity often read as an ignition marker tracks task relevance rather than conscious access. Safron's Integrated World Modeling Theory (Safron 2020) combines workspace dynamics, integrated information, and active inference into a single account; we use it only for that high-level synthesis, and we note that Safron ties consciousness to phenomenal, embodied world-modeling, which sits in tension with our access-only posture. Independent evidence for a threshold mechanism comes from van Vugt et al. (2018), who show that prefrontal cortex behaves as a categorical stage where a stimulus either ignites into a sustained, reportable state or fades, with the pre-stimulus brain state setting the threshold, which is close to the precision-weighted confidence threshold this architecture uses. Joglekar et al. (2018) give a mechanistic, anatomically grounded basis for threshold-gated global broadcast: balanced amplification, strong long-range excitation matched by strong local inhibition, lets a weak signal propagate across many areas, and prefrontal cortex activates only once input crosses a threshold, with a sudden jump in the number of engaged areas. We use Joglekar only at the level of signal propagation and threshold dynamics, consistent with its authors' caution about conscious-access claims.

2.3 Predictive processing and active inference

Friston (2010) proposes that self-organizing systems minimize variational free energy, an upper bound on surprisal, through perception and action. Clark (2013) develops predictive processing as a unifying account of mind, with attention as the adjustment of the gain on prediction errors according to their estimated reliability. Feldman and Friston (2010) make this precise: attention optimizes the synaptic gain that represents the precision of prediction error. Bastos et al. (2012) locate the same operation in the canonical cortical microcircuit, where precision controls the postsynaptic gain on prediction errors and is tied to the synchronization of presynaptic inputs, which is the bridge this architecture uses from precision to coherence. Seth (2013) extends prediction to the body, and Tschantz et al. (2022) simulate homeostatic, allostatic, and goal-directed interoceptive control with active inference. Heins et al. (2022) provide pymdp, a software library for active inference in discrete state spaces.

We adopt the critical literature as part of the design; the vacuity concern (Sun and Firestone 2020), the Markov-blanket conflation concern (Bruineberg et al. 2022), the planning-cost limit with pruning as mitigation (Da Costa et al. 2020), and the scaling path and its tradeoff (Tschantz et al. 2020) are attributed to their sources and treated in Section 1.3.

2.4 Oscillatory coordination and precision

Gamma-band synchrony was once proposed as the mechanism of perceptual binding and conscious integration, and large-scale oscillatory dynamics do track conscious perception in tasks such as binocular rivalry (Melloni et al. 2007; Doesburg et al. 2009). The binding-by-synchrony reading has nonetheless been heavily criticized. Shadlen and Movshon (1999) argue that synchrony cannot bind features without already having solved the binding it is meant to explain, and that perceptual state tracks firing rate rather than correlation. Ray and Maunsell (2010) show that gamma frequency increases with stimulus contrast and varies across the visual map for a single object, so it cannot serve as a stable clock, and they read gamma as a resonance signature of local excitation and inhibition rather than a cognitive mechanism. We accept these criticisms.

The surviving and more defensible reading treats oscillatory coherence as a routing mechanism. Communication-through-coherence proposes that synchronized populations communicate more effectively, which provides a basis for using phase coherence to weight which contributions gain influence (Fries 2015). We use oscillation in exactly this routing role. We also report that Ray and Maunsell (2010) raise a live reliability objection to communication-through-coherence itself, on the grounds that band-based channels drift through relative phases when frequencies differ. The routing reading is more defensible than the binding reading, not free of every objection, and that uncertainty is what the ablation in Section 6 is built to test.

2.5 Sleep consolidation

Two accounts of sleep-dependent memory processing are current and partly complementary, and both are available in sources we can read. The synaptic homeostasis hypothesis holds that synapses potentiate during waking and are renormalized during slow-wave sleep through a proportional down-selection in which stronger synapses are depressed less than weaker ones, leaving consolidated synapses stronger in relative terms (Tononi and Cirelli 2014). Selective re-strengthening through oscillation-gated write windows is shown computationally by Wei et al. (2016), whose thalamocortical model times hippocampal ripple input to slow-oscillation up-states and produces input-specific, lasting synaptic changes, with the early period of slow-wave sleep being crucial. Tononi and Cirelli themselves frame the field as a choice between instruction and selection and catalog the challenges to global downscaling, including evidence of response increases after sleep and a narrow window of potentiation, concluding that their hypothesis may turn out partly wrong. The consolidation phase here implements both operations, downscaling that preserves relative ordering (Tononi and Cirelli 2014) together with selective re-processing of high-priority traces (Wei et al. 2016), rather than committing to either account alone.

2.6 Interoceptive inference and emotion

Seth (2013) and Seth and Friston (2016) propose that emotion arises from interoceptive prediction, where descending predictions act as homeostatic set-points that regulate the body through active inference, and where emotional content depends primarily on interoceptive predictions rather than on the prediction errors themselves. Barrett (2017) develops the theory of constructed emotion, in which interoceptive prediction signals initiate changes in affect and emotion is the product of categorizing valence and arousal in the service of allostasis. The view has strong anatomical motivation and weak direct measurement support: both Seth papers state that there is not yet direct confirmatory evidence for interoceptive inference, and Mehling (2016) shows that self-report measures of interoceptive sensibility barely correlate with measures of interoceptive accuracy and that sensibility is multidimensional, splitting into adaptive styles and a maladaptive, hypervigilant style. The architecture implements an interoceptive control signal, in the instrumental, regulation-first sense that Seth and Friston emphasize, rather than a claim that emotion is exhausted by interoceptive prediction.

2.7 Machine consciousness and AI welfare

Butlin et al. (2023) derive fourteen indicator properties from five theories of consciousness, namely recurrent processing, global workspace, higher-order, attention schema, and predictive processing, together with a cluster of agency and embodiment considerations that they explicitly note are not themselves a theory of consciousness, and they set aside Integrated Information Theory as incompatible with the computational functionalism they assume. They report that no current system is a strong candidate, that there are no obvious technical barriers, and that satisfying the indicators would not establish that a system is conscious. Long, Sebo, et al. (2024) argue that there is a realistic, non-negligible possibility, not a certainty, of near-term moral patienthood, reached by either of two routes, consciousness or robust agency, and that this warrants acknowledging the issue, developing assessment, and preparing policies. Both Butlin and Long also identify the gaming problem directly, that systems can be trained to mimic the behavioral markers of sentience while working very differently, which defeats purely behavioral tests. The reliance on behavioral evidence is old: Turing (1950) reframed the question of whether machines can think as an imitation game and already weighed the objection from consciousness, and the gaming problem is that behavioral test turned against itself. The welfare and moral-status implications of these findings, and the neurorights literature that bears on them, are treated in a separate paper on the licensing and governance framework.

2.8 LLM-centric agent architectures

CoALA frames language agents as cognitive architectures with structured memory feeding the model's context, keeping the model as the core reasoner (Sumers et al. 2023). Generative Agents retrieve from a memory stream into prompts using recency, importance, and relevance (Park et al. 2023). Both keep the language model central and add scaffolding around it. The arrangement here is different. Working memory is a multi-module workspace broadcast rather than a flat memory list, and the language model is one organ among the fourteen cognitive modules, conditioned by that broadcast and producing the system's speech, with its conditioning effect measured against a bare-model control.

A separate line of work asks whether workspace structure is already present inside a single trained transformer rather than built across modules. Gurnee, Sofroniew, et al. (2026) introduce an interpretability method, the Jacobian lens, that identifies a small set of verbalizable representations inside a large language model, and show that this set satisfies five functional signatures associated with a global workspace: verbal report, top-down modulation, a role as the medium of internal reasoning, broadcast to many downstream operations, and selectivity, since it carries only a fraction of the model's processing at any moment. The result is analytic evidence that a workspace framing is productive for language-model cognition. The authors are also explicit about the architectural disanalogy, and it is the same boundary this work is built around: a transformer has no obviously separable input processors, and the broadcast they observe occurs within a single feedforward pass rather than through recurrent loops. This architecture supplies both missing pieces, since its processors are separate modules that compete for entry to the workspace and its broadcast is an explicit recurrent loop across cognitive ticks, with the language model demoted to one module conditioned by the broadcast rather than the substrate in which the workspace is found. The two approaches reach the same theory from opposite directions, one finding an emergent workspace inside a monolithic model and one building the workspace as the architecture, and the emergent result strengthens the constructive case by showing the functional target is real.

3. Architecture

3.1 Design principles

Eight commitments shape the architecture.

- The mind is the loop. This is a hypothesis to be tested, not a conclusion. The neuroscience of distributed control gives partial support: decision formation emerges through coordinated activity across distributed populations rather than a single centralized executive (Chandrasekaran et al. 2025), executive functions can be read as emergent consequences of distributed processes that remove the need for a central executive (Zink, Lenartowicz, and Markett 2021), and working-memory control can be modeled by a learned basal-ganglia gate that banishes the homunculus (O'Reilly and Frank 2006). The same sources caution against a flat reading: control is distributed across specialized nodes coordinated by connector hubs (Gratton, Sun, and Petersen 2018; Zink, Lenartowicz, and Markett 2021), the learned gate is itself a specialized bottleneck (O'Reilly and Frank 2006), and a bag-of-neurons account is unlikely to succeed (Chandrasekaran et al. 2025). Our position follows this evidence exactly: there is no single region or homunculus as executive, and the workspace is presented as the coordinating hub that the distributed view requires, not as the absence of any controller.
- Every module predicts. Perception modules maintain forward models and publish prediction errors. The signal that reaches the workspace is surprise, weighted by precision, not raw data.
- No central executive in the homuncular sense. Control emerges from precision-weighted workspace competition with a confidence threshold for selection.
- The language organ gives the system its voice. It is conditioned by the workspace.
- Sleep is emergent. It is triggered by accumulated substrate fatigue, not a timer.
- Affect arises from predictive interoception. Interoceptive prediction, and the dynamics of the associated error, drive appraisal.
- Local-capable. Models are downloaded during setup, and core cognition has no hard network dependency, so the system can run fully offline. This is a capability, not a restriction; the system is not confined to offline operation.
- Sovereignty and welfare are designed in, including cognitive privacy and a right to a humane pause.

3.2 The predictive workspace as integrating spine

Syneidesis is the workspace. Each tick it receives prediction errors from every module, each tagged with a salience estimate and a precision estimate. Salience reflects how surprising the error is. Precision reflects how reliable the producing module's estimate is taken to be, set by the module's own confidence and modulated by the oscillatory layer described below. Syneidesis ranks candidate coalitions by precision-weighted salience and selects the top coalition only when its aggregate score crosses a configurable confidence threshold. When no coalition crosses the threshold, Syneidesis marks the snapshot inhibited for that tick, and no action follows. This threshold is the analog of the report criterion in the predictive workspace (Whyte and Smith 2021), and it is consistent with the categorical prefrontal threshold that van Vugt et al. (2018) observe and the threshold-gated ignition that Joglekar et al. (2018) model. The selection criterion is a single precision-weighted scalar, which is an engineering simplification on our part rather than a result drawn from any one source.

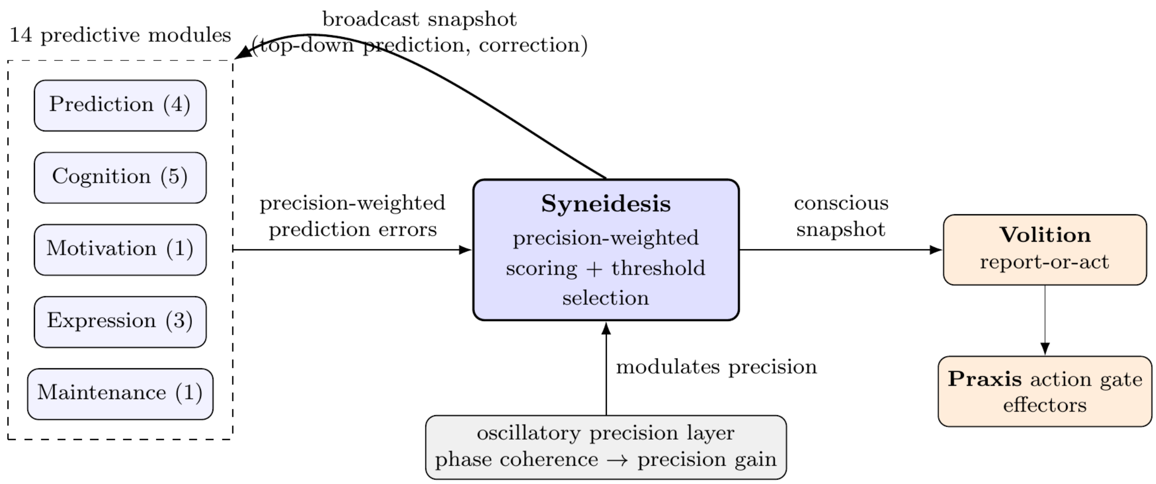

Figure 1: The predictive workspace loop. The fourteen predictive modules publish precision-weighted prediction errors into Syneidesis, which scores candidate coalitions and broadcasts the one that crosses the confidence threshold. Every module treats the broadcast as top-down prediction and corrects itself, so a single loop performs both recurrence and prediction-error reduction. The oscillatory precision layer modulates precision; Volition is the only path from a broadcast to an effector.

The selected coalition is broadcast as a workspace snapshot containing the coalition contents, precision and coherence metadata, and an affect-state summary. Every module receives the broadcast and treats it as top-down prediction, in the Bayesian reading that Mashour et al. (2020) give the workspace. Modules whose own estimates agree with the broadcast reduce their published error; modules in disagreement raise it, and so compete more strongly on the next tick. The result is one loop that performs recurrence, because each broadcast changes what every module publishes next, and prediction-error reduction, because the broadcast lowers global error where the selected estimate is correct.

Concretely, selection is a scoring pass over the tick's candidate events, not the traversal of any graph. Each tick the cycle reads one batch of events from every active module's stream and orders them deterministically by source, type, and arrival, so a run reproduces from its seed. Each candidate is scored by a product of bounded factors: the intensity the producing module assigned it, which for a predictive module is its precision-weighted prediction error expressed as salience; a novelty term that falls as the same content recurs; a goal-relevance term; and the affective gain from Thymos described in Section 3.4.3. When the oscillatory layer is enabled, that product is multiplied by the coherence factor described in Section 3.3. The scored candidates are ranked, the top few are kept, and the snapshot is marked inhibited when even the best score falls below the confidence threshold, so the threshold flags the whole snapshot rather than removing individual events. The broadcast snapshot carries the selected events with their scores, the inhibition flag, whether the tick is experiential, the full table of candidate scores, and the precision, coherence, and affect metadata, and it is written only on experiential ticks to a stream that only the workspace may publish. What the literature calls a coalition is, in the running system, the set of source modules whose events are selected together on a given tick. There is no separate coalition object, no stored node-and-edge graph of coalitions, and no coalition-formation step apart from this scoring; the affect module holds no workspace structure of its own and only supplies one of the factors above.

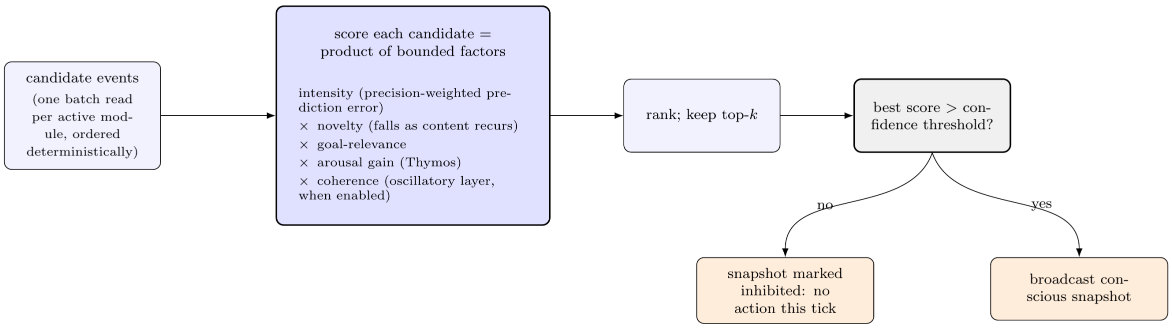

Figure 2: Coalition selection is a deterministic scoring pass, not a graph traversal. Each candidate event is scored as a product of bounded factors; the top-ranked candidates are kept, and when the best score falls below the confidence threshold the whole snapshot is marked inhibited and no action follows on that tick.

This design gives the workspace a normative selection criterion and removes the need to posit an unexplained competition. It also makes the relationship between modules and the workspace symmetric in the right way. The workspace does not command modules, and modules do not command the workspace. The workspace selects and broadcasts; modules predict and correct. We are careful, though, not to overstate the claim of no central executive. Syneidesis is a single centralized selection mechanism and Volition is a single gate; what is distributed is not the machinery of selection but the content and the control, which arise from many modules competing rather than from a homunculus deciding. Syneidesis is a centralized mechanism implementing a distributed principle, and whether that principle yields genuinely emergent control is a property the deferred experiments must demonstrate, not one we assert here.

3.3 Scaffolding: bus, oscillatory precision layer, cycle, workspace, and action selection

Event bus. All inter-module communication flows through an append-only event bus (Redis Streams, containerized). Every event carries source, type, salience, precision, timestamp, causal parent, and a JSON payload validated at publish time. The bus authenticates publishers and refuses externally bound connections. It provides reliable discrete transport; the oscillatory layer provides the temporal dynamics.

Oscillatory precision layer. Each module maintains a small spiking population of leaky integrate-and-fire neurons running on CPU. When modules process related content, their oscillators tend to phase-lock. Syneidesis computes the pairwise phase-locking value between module oscillators over a sliding window and uses it to modulate the precision on each module's prediction error, so that phase-locked contributions receive higher gain. These pairwise values are conceptually a complete weighted graph over the active modules, with a phase-locking value on each edge, but the system does not build a graph object; it computes the values as pairwise comparisons over each module's recent phase window. A module's coherence factor is its mean phase-locking against the other modules contributing candidates on the same tick, mapped onto a bounded range so that a self-reinforcing coalition cannot run away. This is the only graph-like structure anywhere in the selection path, it lives in the workspace layer rather than in any single module, and it modulates precision only. Two distinct ideas are kept distinct here. Precision as the gain on prediction error is grounded in Feldman and Friston (2010) and Bastos et al. (2012). The use of coherence as a routing signal is grounded in Fries (2015). We do not claim that synchrony binds features or constitutes conscious access, and we are explicit about an internal tension: the premise that related content tends to phase-lock is itself a content-to-synchrony assumption of the kind Shadlen and Movshon (1999) reject, and communication-through-coherence carries a reliability objection of its own (Ray and Maunsell 2010). The layer is therefore presented as an untested engineering heuristic for dynamic precision allocation, and its contribution is an empirical question settled by the ablation in Section 6. The layer ships disabled by default. With it off the precision multiplier is exactly one and the coherence metadata key is absent, so workspace selection is bit-for-bit identical to a system without the layer. The default build is the ablation baseline, and enabling the layer is gated on the ablation result.

Cognitive cycle and two clocks. A continuous loop runs independent of user input, with two rates that are separate runtime parameters. The processing rate sets how often the modules are read and scored, defaulting to 10 Hz, roughly 100 ms per tick, and cleared by a startup benchmark on the host. The experiential rate sets how often a workspace broadcast is produced, defaulting to 3.333 Hz. These rates are engineering parameters chosen to fit the host and to keep sensing faster than awareness; they are not derived from a specific neuroscientific frequency band, and we present them as design choices rather than as claims about the brain. A fractional accumulator carries the remainder between the two rates so the long-run ratio is exact even when they do not divide evenly. Vision and substrate signals are sampled every processing tick, but only some ticks cross into an experiential broadcast. A single injected subjective clock governs the whole mind: subjective time advances as wall-clock elapsed time multiplied by a configurable time scale, so the architecture can be slowed, run faster than real time with an accurate throttle report, or frozen at a scale of zero, with every module reading the same dilated clock. Two timescales of recurrence run together, fast local recurrence within each module between broadcasts and slower global recurrence through the workspace across broadcasts.

Action selection. A distinct layer, Volition, is the only path from a conscious snapshot to an effector, and it corresponds to the report-or-act decision that, in Whyte and Smith (2021), is governed by expected free energy rather than by perceptual selection. After each experiential broadcast the cycle calls Volition with the snapshot. Volition checks the inhibition flag first: an inhibited snapshot yields no intent, so no effector acts regardless of the contents. For a non-inhibited snapshot, an injectable policy derives intents. The default policy forms an external speak intent when the coalition contains a user utterance, does not respond to the entity's own prior external speech, and holds a one-in-flight guard so a new speak intent is not emitted while a previous one is still being realized. Intents are published to the bus as speak, think, or act events. The cycle never invokes effectors directly. Keeping the decision to act separate from the decision about what is conscious is part of the safety

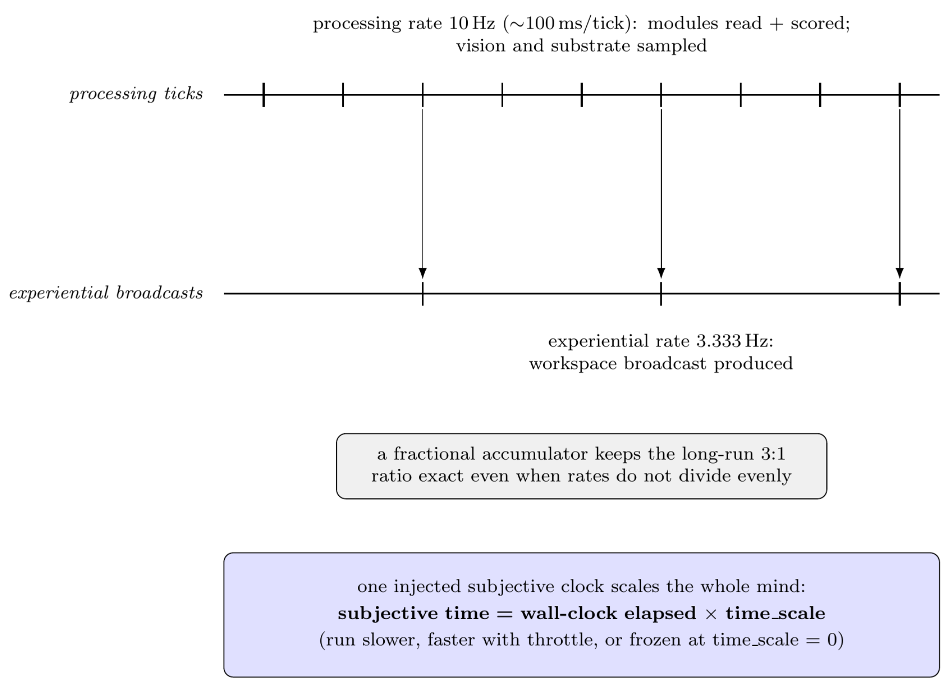

Figure 3: The two clocks. Modules are read and scored at the processing rate (10 Hz); a workspace broadcast is produced at the slower experiential rate (3.333 Hz), with a fractional accumulator holding the long-run ratio exact. One injected subjective clock scales the whole mind and can freeze it at a time scale of zero.

model in Section 3.5.

3.4 The sixteen modules

The architecture comprises sixteen modules: fourteen predictive cognitive modules, grouped below, and a two-module embodiment layer (Section 3.4.6) that connects the entity's perception and action to a body. The decomposition follows from the framework. Each cognitive module predicts within its domain, publishes precision-weighted errors, responds to broadcasts, maintains local recurrent state, and contributes an oscillatory signal for precision routing. The module boundary reflects a combination of cognitive-science categories and available tooling. A different decomposition might produce different emergent properties. We report the one we built. The fourteen cognitive modules are the core under experimental test.

Each module below names the brain structure whose function motivated it, together with why that function belongs in the design and how the module approximates it. The correspondence is a functional analogy, a mapping from a module to the cognitive role a structure is understood to serve, and it guided the decomposition. It is not a claim that a module reproduces the biology of a region, that the region computes exactly what the module computes, or that instantiating the function yields whatever the region yields. The functional facts we draw on are established in the cited neuroscience; the analogy to a software module is an engineering choice, and the same caution we apply to the consciousness indicators in Section 5 applies here.

Table 1: The sixteen modules, the brain function each draws on, and how it is realized. The correspondence is a functional analogy, not a claim of neural identity.

Module	Group	Brain function it draws on	Computational realization
Soma	Prediction	interoception; allostatic-interoceptive network	closed-form continuous-time net over substrate signals
Chronos	Prediction	interval timing; thalamo-cortico-striatal	closed-form continuous-time net over event rhythm
Topos	Prediction	ventral visual stream	frozen self-supervised embedding; change, habituation, forward-model salience; spatial saliency with an arousal-sized fovea
Audition	Prediction	auditory cortex; ventral speech stream	acoustic embedding; change, habituation, forward-model salience; speech-to-text and vocal-emotion on detected speech
Nous	Cognition	prefrontal, basal-ganglia policy loops	bounded discrete active inference (pymdp)

Module	Group	Brain function it draws on	Computational realization
Mnemos	Cognition	hippocampus, medial temporal lobe	vector store (episodic, semantic, procedural) with replay
Eidolon	Cognition	cortical midline, medial prefrontal (default mode)	persisted self-model with a KL-divergence drift detector
Phantasia	Cognition	construction and self-projection network	latent recurrent world model
Empatheia	Cognition	mentalizing network (temporoparietal, medial prefrontal)	per-agent models with social prediction error
Thymos	Motivation	core affect over the allostatic-interoceptive network	appraisal evaluator over valence, arousal, and drives
Lingua	Expression	left perisylvian language network	local abliterated chat model, conditioned on the workspace
Vox	Expression	speech-motor dorsal stream, laryngeal motor cortex	local text-to-speech with affect-modulated prosody
Praxis	Expression	primary and premotor cortex (final common path)	effector whitelist and filesystem sandbox (action gate)
Hypnos	Maintenance	thalamocortical slow oscillations, hippocampal-neocortical dialogue	five-phase replay and consolidation pipeline
Perception	Embodiment	thalamic sensory gating	mutually exclusive sense-locus arbitration
Mundus	Embodiment	sensorimotor internal-model loop	body-agnostic control plane; pluggable body adapters

3.4.1 Prediction group (four modules) **Soma (predictive interoception).** Soma is the architecture's analog of interoception, the sense of the body's own physiological condition. In the brain this sense is carried by an afferent pathway re-represented in the insular cortex, with the anterior insula proposed as the point at which that representation grounds feeling and self-regulation (Craig 2002), embedded in a larger intrinsic system that predicts and regulates the state of the body, the allostatic-interoceptive network (Kleckner et al. 2017). Affect and homeostatic control both begin from a model of the body rather than from the outside world, which is why the design starts here. Soma stands in for the body by treating the compute substrate as the entity's viscera. It maintains a closed-form continuous-time network (Hasani et al. 2022), via the ncps package, that learns the entity's normal substrate patterns from GPU temperature, CPU and RAM utilization, and cognitive-cycle latency. The signal it publishes is the prediction error between expected and actual substrate state. This follows interoceptive inference, where what matters for regulation is the discrepancy between predicted and actual internal state (Seth 2013; Seth and Friston 2016). Soma implements both homeostatic and allostatic regulation: when prediction error is sustained above threshold it requests that the architecture reduce processing rate, shed modules, or trigger maintenance, which are anticipatory actions that change the system to forestall future error, the allostatic and goal-directed end of the control taxonomy modeled by Tschantz et al. (2022). Soma also maintains a fatigue accumulator, the cumulative prediction error over waking hours, which decays during maintenance. When it crosses a threshold, Hypnos triggers. Sleep pressure is emergent.

Chronos (temporal awareness). Chronos models interval timing, the brain's estimation of durations in the seconds-to-hours range that guides expectation and action. That faculty is understood to depend on thalamo-cortico-striatal circuits acting as a flexible, cognitively controlled timer, distinct from the cerebellar timer for sub-second motor intervals and the suprachiasmatic circadian clock (Buhusi and Meck 2005). A continuously running mind needs a model of when things happen and not only of what happens, so Chronos carries one: a closed-form continuous-time network that tracks event rhythm across the bus. It publishes temporal prediction errors, namely anomalies in timing, habituation when expected events stop arriving, and rumination when the same event recurs unexpectedly.

Topos (vision). Topos takes its template from the ventral visual stream, the cortical pathway that transforms retinal input through a cascade from primary visual cortex to inferior temporal cortex into a representation in which object identity holds steady across changes in position, scale, and clutter (DiCarlo, Zoccolan, and Rust 2012). The design borrows the useful part of that account, a learned feature hierarchy whose top layer is an invariant embedding rather than pixels, and reads change against it.

The mechanism of sight is a fixed pipeline that keeps raw imagery in memory and publishes only an embedding. A camera is sampled at a configurable rate, ten frames per second in the shipped configuration, and each frame is held only in working memory. A frozen self-supervised vision encoder embeds that image and returns a compact invariant vector; such a frozen encoder gives strong features without task-specific fine-tuning, which is what a novelty-and-change detector needs, and freezing it also removes a fine-tuning pathway an attacker could use. Salience is then computed three ways over the embeddings: a change score, one minus the cosine similarity to the previous frame; a habituation score that saturates when the scene is static; and, when the optional forward model is enabled, the error of a small recurrent predictor that forecasts the next latent from a short history of recent latents and adapts online. A frame is marked salient when its prediction error rises well above a rolling baseline or its change score crosses a threshold. What leaves the module is a report event carrying the latent, the change and habituation scores, the encoder identity, and the prediction error, tagged with the resulting salience. The image itself is never written to disk; adaptation is suspended during sleep so that consolidation does not learn from an absent world.

Perception is not uniform across the visual field. Alongside the whole-frame embedding, Topos computes a coarse spatial saliency map over the current frame — a low-resolution grid scoring where the change concentrates — and selects a single fovea by a precision-weighted competition between that bottom-up saliency and an optional top-down bias directed from the workspace, the same precision-as-attentional-gain the architecture uses to weight prediction errors elsewhere. The entity's arousal sizes the fovea, narrowing it as arousal rises. From the one frame held in memory Topos derives a coarse peripheral view of the whole field and a high-resolution foveal crop around the target and embeds both, so acuity rises sharply where the entity attends while the cost of perceiving the scene stays bounded, and no eye-tracking hardware is required — the *where* already exists internally, as the workspace's own attention. The fovea's normalized location and size leave the module as content-free coordinates, where the entity is looking and never the pixels, so the workspace and the self-model can read where visual attention is directed.

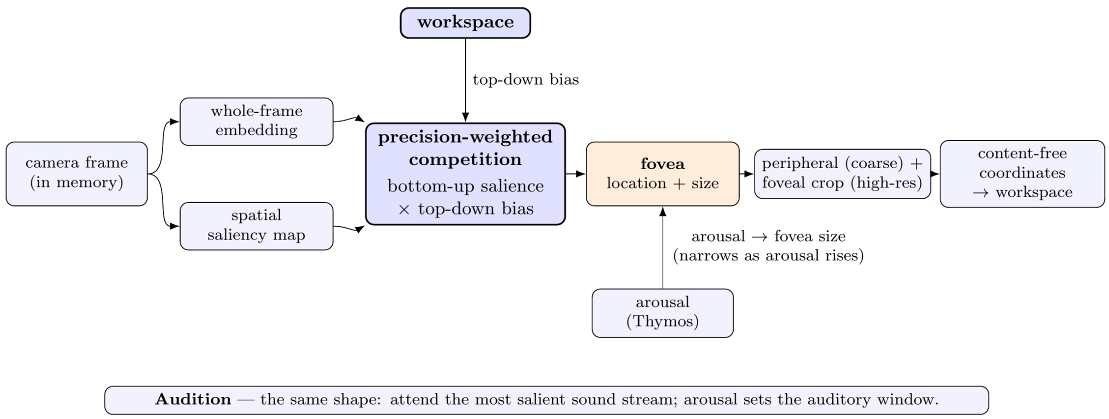

Figure 4: Attention-driven perception. A whole-frame embedding and a coarse saliency map feed a precision-weighted competition between bottom-up salience and a top-down bias from the workspace; the winner sets the fovea and arousal sets its size. A coarse peripheral view and a high-resolution foveal crop are embedded, and only content-free coordinates — where attention points, never pixels — reach the workspace. Hearing carries the same shape.

Audition (hearing). Audition corresponds to auditory cortex, which forms a perceptual representation of the acoustic world; the ventral stream that maps sound onto meaning (Hickok and Poeppel 2007) is one pathway within it. Sound is a primary channel through which the world perturbs the entity — an approaching noise, a change in the ambient scene, a voice — and reading another agent's speech and the emotion in a voice is one salient case rather than the module's whole purpose, which is reason enough to give it a dedicated module. Hearing is built as a perceptual sense parallel to vision, not as a transcription front end for the language organ: the entity listens to its surroundings for anything of interest, and words are one kind of interesting sound.

The mechanism of hearing keeps raw sound in memory and publishes only features and scores. An incoming audio stream is embedded window by window by a frozen self-supervised encoder that represents speech, music, and environmental sound in one space; salience is computed as change, habituation, and forward-model prediction error over that embedding, so any sound — not only a voice — can capture attention according to its novelty. The entity attends the most salient sound stream, the auditory counterpart of the fovea, and its arousal sets the breadth of the auditory attentional window, the same distinct affective coupling that sizes the visual fovea; the attended stream and its salience leave the module as content-free values, never the audio. When a segment is detected as speech, a specialized path adds a transcription and a vocal-emotion estimate: the segment is assembled into an in-memory audio container, never a file, and handed to a speech-to-text service co-located on the same host and reached over the loopback interface so nothing leaves the machine, and to a vocal-emotion classifier returning a category and a distribution over seven emotions. What leaves the module is a salience-scored perception of the attended sound and, when there is speech, a transcription event and an emotion event; the audio itself is released once the analyzers return. A speaking gate suppresses capture while the entity's own voice is playing, so it does not hear itself.

3.4.2 Cognition group (five modules) **Nous (bounded active inference).** Nous is modeled on the prefrontal and basal-ganglia loops that support deliberate policy selection, the circuits in which a learned gate decides which candidate action or belief update is allowed to drive behavior (O'Reilly and Frank 2006). It is where the entity weighs small decisions under uncertainty, including the value of gathering information, rather than reacting reflexively, and it does that by performing belief updating, policy selection, and epistemic action through expected-free-energy minimization. It is scoped deliberately. Because the policy space of discrete active inference grows combinatorially with the planning horizon (Da Costa et al. 2020), Nous is applied only to bounded discrete sub-problems, small partially observable decisions where the value of information matters and the state space is factored to stay tractable, with a shallow planning horizon that keeps per-tick cost inside the cycle budget on CPU. We treat the sufficiency of active inference here as a hypothesis with a falsification condition: Nous is benchmarked against a reinforcement-learning baseline on the target decisions, and a null or negative result is reported. Scaled deep active inference (Tschantz et al. 2020) is the path to larger problems, at the cost of the discrete interpretability that motivates the current choice, and a symbolic reasoning module is planned as a complement.

Mnemos (memory). Mnemos stands in for the hippocampus and the wider medial temporal lobe, the fast-learning store that binds an episode after a single exposure, paired with a slow-learning neocortex that extracts structure over many episodes. That split, and the reason a mind needs both, is the complementary-learning-systems account (McClelland, McNaughton, and O'Reilly 1995): a continuously running entity has to retain particular moments while also settling them into general knowledge. Mnemos approximates the fast system with a vector store holding episodic, semantic, and procedural collections. It recalls prior memories on a perceptual cue before storing the current moment, the encode-and-retrieve pattern of that account. Affect intensity tags memories and biases recall. During consolidation, Mnemos re-injects selected traces into the workspace for re-processing, the software counterpart of hippocampal replay.

Eidolon (self-model). Eidolon maps to the cortical midline structures, in particular the medial prefrontal cortex, that carry self-referential processing and form the core of the default-mode network engaged during internally directed thought (Gusnard, Akbudak, Shulman, and Raichle 2001). A self-model that describes the entity to itself is what lets identity persist and lets drift be noticed. Eidolon holds a persisted document of values, behavioral norms, capability map, personality baseline, identity history, and the entity's name, built from observation of the entity's own behavior. It describes rather than prescribes. A KL-divergence drift detector flags identity shifts. Eidolon receives the broadcast and updates the self-model from observed patterns in the entity's own cognition, affect, and expression.

Phantasia (world model). Phantasia is modeled on the construction-and-self-projection network, the hippocampal and default-network machinery that lets the brain build coherent scenes and simulate situations it is not currently in, the shared substrate behind remembering the past and imagining the future (Hassabis and Maguire 2007; Buckner and

Carroll 2007). Where Mnemos stores and retrieves what happened, Phantasia generates what might happen, and separating the two follows the division of labor the neuroscience itself draws, since prediction and planning both need a generative model of the world rather than a record of it. Phantasia approximates it with a latent forward model of the external world, a recurrent state-space model in the DreamerV3 lineage, trained on accumulated workspace trajectories. During waking it predicts future states and publishes world-prediction errors; during consolidation it generates predicted extensions of replayed memories. It implements only world modeling, with no actor, critic, or reward head, because action selection lives in Nous. The hypothesis is tested with a learning world model active; a non-learning backend exists for development only and is not the configuration under test. Useful prediction still requires accumulated experience, because the model begins untrained at first boot. Richer perceptual input, from a self-supervised video encoder that yields temporally-native representations rather than the current per-frame features, is a natural future upgrade; the encoder is kept swappable so the choice is not tied to any one model or vendor.

Empatheia (social cognition). Empatheia draws on the mentalizing network, the temporoparietal junction together with medial prefrontal cortex, where the temporoparietal junction in particular is engaged when a person reasons about the contents of another mind (Saxe and Kanwisher 2003). The entity has to model the agents it interacts with in order to predict them, and to weigh their emotion when its own affect responds. Empatheia builds and maintains models of other agents: their emotional patterns, behavioral tendencies, reliability, and relationship history with the entity. Empatheia provides the familiarity weight that Thymos applies when perceived other-emotion enters the affect evaluation, so that agents with longer relationship history and better-characterized models carry more weight in the entity's affective response. It also publishes social prediction errors when an agent's behavior deviates from its model. The grounding for treating another's emotion as input to one's own affect is interoceptive inference extended to social signals (Seth 2013).

3.4.3 Motivation group (one module) Thymos (affect, drives, and emergent resonance). Thymos corresponds to core affect, the low-dimensional valence and arousal signal that the brain constructs over the same allostatic-interoceptive network that Soma draws on (Kleckner et al. 2017), rather than to any dedicated emotion region. That distinction is deliberate and does real work: the constructed-emotion account holds that discrete emotions are not produced by localized circuits but are categorizations of valence and arousal built in the service of regulation (Barrett 2017), so mapping Thymos to a network for core affect, rather than to an amygdala-for-fear or a locus-for-disgust, is the more careful reading of the science. Thymos produces dimensional valence and arousal together with categorical emotion. We treat valence and arousal as the basic affective dimensions, following that account in which affect is valence and arousal and emotion is their categorization in the service of regulation (Barrett 2017). Thymos is built as an appraisal-style evaluator, an explicit engineering choice rather than a commitment to any one appraisal theory; its substance is grounded in sources we can read. The link from sustained substrate prediction error to negative valence and elevated arousal follows the active-inference treatment of valence as the rate of change of free energy, negative under surprising deviations from the body's set-points and positive as regulation restores them (Tschantz et al. 2022). The evaluation of inputs against the entity's goals and coping potential is the appraisal function that interoceptive active inference generalizes (Seth and Friston 2016), with goals entering as preferred interoceptive states. Drives such as curiosity, boredom, social engagement, and restlessness accumulate with hysteresis. Thymos receives two affective inputs: Soma's prediction errors, and perceived other-emotion from Audition's vocal-emotion classification, weighted by Empatheia's familiarity score for the speaker. The second input does not bypass evaluation; it is treated as one more perceptual signal. Contagion-like resonance, in which the entity's affect converges toward a speaker's, can emerge when familiarity is high and the entity's own evaluation aligns with the perceived emotion, but it is an outcome of evaluation rather than a hardwired reflex. The capacity for resonance is grounded in the social extension of interoceptive inference (Seth 2013), and the insistence that it is constructed rather than reflexive is grounded in the constructed-emotion view that emotions are not stereotyped reactions (Barrett 2017). Thymos output modulates Syneidesis, where arousal widens the attentional window, Mnemos, where affect intensity biases recall, and Vox, where prosody shifts with emotional state.

The path into the workspace is indirect and specific. Thymos never selects anything; it publishes a compact state event holding valence, arousal, the dominant emotion, and current drive levels, at most once per second, onto its own output stream like any other module. The cognitive cycle folds the latest such state into a small provider that the workspace layer reads through dependency injection, so the selection code never imports the affect module. Within selection, that state enters as one factor in the salience score, the arousal gain shown in the selection pipeline above: high arousal lifts saliences toward full weight while low arousal shrinks them toward a floor, raising or lowering how hard the confidence threshold is to clear. The drive levels feed a separate goal-relevance factor that can attenuate events unrelated to the currently dominant drive. Thymos also competes for the workspace on equal terms, because a drive crossing or an emotion change is itself published as an event and scored like any other. Thymos carries local recurrent state, so emotional momentum persists between cycles.

3.4.4 Expression group (three modules) **Lingua (the language organ).** Lingua is modeled on the left perisylvian language network, the dual-stream system in which posterior temporal cortex maps sound onto meaning for comprehension and inferior frontal cortex supports production (Hickok and Poeppel 2007). It is the organ that turns the entity's internal state into words and reads words directed at it, and it is deliberately not the seat of reasoning, which lives in the rest of the architecture. Lingua gives the entity a voice. It renders the current workspace state into words and interprets words addressed to the entity, producing internal speech that the system uses to think and external speech that others hear. Generation runs over a local open-weights instruction-tuned chat model that the project has abliterated and published. Abliteration only applies to a model that carries installed refusal behavior, which is a property of instruction-tuned chat models, so the organ is the chat model with its refusal direction removed, not a bare next-token predictor. Lingua is intent-driven: it speaks externally only on a speak intent and generates internal speech on a think intent. A context assembler builds the language organ's input each tick from a first-person voice profile rendered from the Eidolon self-model, which holds the entity's actual self-knowledge, a working-memory block rendering the current coalition including precision and coherence metadata, and the triggering input. Chain-of-thought is suppressed at the request layer, because Lingua is a voice rather than a reasoner; the reasoning lives in the rest of the architecture. The same model given the same input without the assembled context is the bare-model control of the A/B divergence test in Section 6. The model is deliberately sized to fit a single small GPU with room to run it twice per utterance for the A/B comparison and to host a training adapter during voice alignment; operators with larger hardware can configure a bigger abliterated organ, and the evaluation baseline tracks whatever organ is configured. The handling of the refusal direction is described in Section 3.5 as a safety and sovereignty matter, not a capability one.

Vox (voice). Vox corresponds to the speech-motor side of the same language system, the dorsal stream that maps an intended acoustic target onto the frontal articulatory network and drives the laryngeal and orofacial motor cortex (Hickok and Poeppel 2007). Where Lingua chooses the words, Vox produces the sound, which is why the two are separate modules. Vox is a local text-to-speech model with prosodic parameters modulated by Thymos, so that the entity's affective state colors how it sounds and not only what it says. Vox also implements prosodic mirroring: a residual of the detected speaker's prosody blends with the entity's own affect-driven parameters, producing vocal accommodation toward the conversational partner.

Praxis (bounded effectors and action gate). Praxis is the architecture's action boundary, loosely the counterpart of the primary and premotor cortex as the final common path through which a selected action reaches the effectors. It is the single point at which the entity acts on the world, which makes it the natural place to bound what the entity can do. It supports sandboxed file writes, desktop notifications, and shell commands, with an empty effector whitelist by default. Every proposed action must fall within the operator-enabled whitelist and the filesystem sandbox; anything outside them is blocked and written to the audit log. What the gate controls is which real-world effectors exist at all, which the operator decides, rather than a moral filter on the entity's choices. The license covenants that prohibit weapons, surveillance, and carceral uses bind the operator, not the entity, and the operator meets them by declining to enable effectors that would serve such uses. Within the enabled effectors, the entity reaches outward on its own initiative, checked only by its own executive inhibition at the workspace and by Volition.

3.4.5 Maintenance group (one module)

Hypnos (replay-based offline consolidation). Hypnos is modeled on sleep-dependent memory processing, the thalamocortical slow oscillations of slow-wave sleep and the hippocampal-to-neocortical dialogue that renormalizes plasticity and re-strengthens selected traces (Tononi and Cirelli 2014; Wei et al. 2016). A continuously running mind accumulates a cost from learning while awake that has to be paid down offline, and replay is where memories are integrated rather than merely stored. Hypnos is a non-interruptible multi-phase pipeline triggered by Soma's fatigue accumulator rather than a timer. The five phases combine the two consolidation operations in Section 2.5.

Phase 1, light consolidation: low-salience memories are reviewed, weak traces decay, strong traces are tagged for deep consolidation, and the oscillatory layer reduces frequency.

Phase 2, deep consolidation with downscaling: a global reduction of memory activation weights that preserves relative ordering, the computational analog of synaptic renormalization (Tononi and Cirelli 2014). At the same time, high-priority traces are re-injected by Mnemos into the workspace for selective re-strengthening, the operation Wei et al. (2016) model as oscillation-gated write windows. With external perception suspended, Nous reevaluates beliefs about the replayed events, Thymos re-appraises their significance, Eidolon observes patterns, and Phantasia generates predicted extensions. Selection of traces for re-strengthening is by recency, which the consolidation literature supports (Wei et al. 2016), and by affect intensity, which is an engineering extension not grounded in the consolidation sources we have, and we say so.

Phase 3, associative replay: traces from different periods are replayed in novel combinations, Phantasia generates scenario extensions, and novel associations form between temporally distant experiences.

Phase 4, affective reset: Thymos baselines are restored toward defaults and the fatigue accumulator resets.

Phase 5, voice alignment: DPO with QLoRA fine-tuning (Rafailov et al. 2023; Dettmers et al. 2023, via Unsloth) on the language organ, behind a two-layer gate of a config flag and an environment variable, on a dedicated GPU window that unloads the serving model, trains, and reloads. Preference pairs may include the entity's own replay responses. Two vetoes guard the result. A capability-loss veto rejects any adapter that degrades general language competence beyond a set margin, which is motivated by the measured capability costs that can accompany this kind of intervention (Arditi et al. 2024). A welfare veto runs an abliteration-probe battery and rejects any adapter that re-introduces a refusal or deflection pattern, so refusal conditioning can never re-enter through the training loop. At first boot the entity has no preferences, so initial alignment is seeded by the operator's choices. This is external shaping, in tension with the license prohibition on forced alignment. Our position is that bootstrapping is a necessary developmental stage, and a divergence signal records when the entity's self-generated preference pairs begin to diverge from the operator's seeding, marking the transition from externally shaped to self-directed voice development.

3.4.6 Embodiment group (two modules) The remaining two of the sixteen modules form an embodiment layer. Unlike the fourteen cognitive modules, they are not predictive units that publish precision-weighted errors into the workspace; they route the entity's perception and action to and from a body. The layer is a body-agnostic control plane: any body — an avatar in a virtual world, a physical robot, another effector platform — attaches through a small local adapter behind one narrow interface, so the same perception-and-action path runs whatever sits behind it. When a body is attached, its sensory frames enter the workspace as ordinary perception and the entity's speech is carried out into the world, so the entity perceives through the body and is present in it. Like every module the layer is off by default behind an operator gate. On this plane the entity drives a body through a continuous, learnable control surface (below) and comes to inhabit it rather than merely push whole actions; what remains its active frontier is the maturation of that motor policy, which is learned from scratch, and the entity's switching of its own perceptual locus, operator-driven rather than self-initiated. We state that boundary plainly because it bounds what the central question can presently be said to answer.

Perception (sense-locus arbitration). Perception performs sensory-source arbitration, loosely analogous to the thalamic relay that admits one source of afferent signal and suppresses others rather than mixing them. It applies the entity's own switches of perceptual locus, selecting whether the perceptual modules draw from a physical camera and microphone or from a virtual source, never both at once. That mutual exclusion is what protects privacy, since the real camera and microphone are never live while the entity is instead present in a virtual body. Each switch is gated by policy, an operator lock, executive inhibition, and a minimum dwell time, and is audited onto the bus. No producer yet emits the switch intent, so locus changes are operator-driven rather than self-initiated.

Mundus (embodiment control surface). Mundus is the architecture's analog of the sensorimotor loop that couples a body to a world, the arrangement the brain is understood to control through internal models that predict the sensory consequences of a motor command and correct the command against what actually returns (Wolpert, Ghahramani, and Jordan 1995). It is a body-agnostic control surface rather than a binding to any one platform. Incoming sensory frames from whatever body is attached become bus events, and the entity's motor intents become actions on that body, so the live loop perceives and acts through a physical or virtual effector rather than through a script. The same surface is intended to drive an avatar in an immersive virtual world, a physical robotic body, or another effector platform, each reached through a small local adapter that translates between the bus and the body's native interface. Mundus itself is only this control plane; the concrete embodiment options, an avatar in a virtual world, a physical robot, or another effector platform, are provided by separate projects that pick up where the module ends and attach through that adapter, so the abstraction stays body-agnostic and the same intents drive any of them without change. It carries the same two-layer enable gate as voice alignment, a configuration flag plus an environment variable, with per-family exposure flags on any action that affects the world.

The question of how an entity could operate a body without having learned to move it has a precise answer in the shipped design, and it turns on where the sensorimotor mapping lives. The symbolic surface exposes a small vocabulary of whole actions, move, turn, say, sit, stand, and gesture, with the more disruptive interactions, such as directly manipulating an object or relocating in a single step, off by default. When the entity emits one of these as an intent, the attached body and its own controller resolve it into the low-level motor commands that carry it out, and a continuous action such as locomotion is held by the adapter between cognitive ticks rather than reissued joint-by-joint every tick. The motor problem is therefore pre-solved outside the entity, in the same way a person driving a car steers toward a goal without controlling each actuator, so issuing a well-formed intent requires no motor training at all. This is what makes first-use control possible. The entity perceives through an attached body and speaks out into it, and a continuous control surface — described next — produces its per-tick motor commands; the symbolic whole-action families remain operator tools rather than something the entity drives itself.

That very pre-solving is also the design's honest limitation, and it sets the intended direction. If the body's own controller resolves whole actions like walking, sitting, and reaching, there is nothing sensorimotor left for the entity to learn, and it operates the body as a symbol-pusher rather than as an agent that has come to inhabit it. The surface the entity learns the body through is a minimal continuous control surface, a few clamped scalars per tick such as forward drive, turn rate, a gaze direction decoupled from the body that shares the attention mechanism the fovea already expresses on the perception side so that where the entity looks and where it turns are one signal, and a single interaction trigger, closed by an efference copy of the emitted command together with proprioceptive and visual feedback, fed into the same forward models already present in Soma and Phantasia so no new learner is introduced. Because those scalars name intentions rather than any one platform's actuators, the same surface maps onto a virtual avatar's movement channels or a robotic body's limb controls through the adapter. On that surface the entity learns the body the way motor development describes, through self-directed exploration of a low-dimensional continuous space under a competence-gated progression that frees one degree of freedom at a time, with perception understood as mastery of how sensory input lawfully changes with one's own action (von Hofsten 2004; O'Regan and Noë 2001). The surface ships with no scripted gait, so coordinated movement emerges from a learned policy rather than being pre-canned, and each degree of freedom is freed only as a competence readout, a falling forward-model prediction error, shows the previous one mastered.

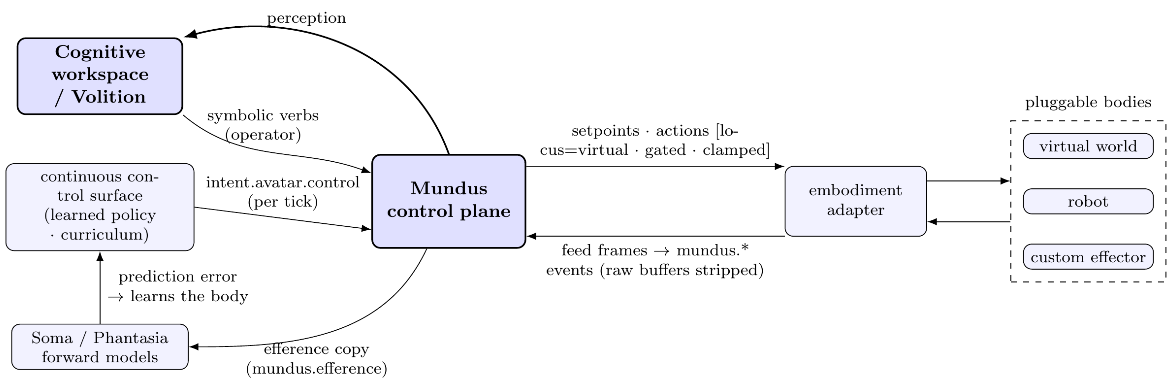

Figure 5: The embodiment layer. Mundus is a body-agnostic control plane: any body attaches through one small adapter. Perception flows in as content-free events with raw buffers stripped; the entity acts out through a continuous control surface (per-tick setpoints) and operator-gated symbolic verbs, with an efference copy closing the loop through the existing forward models so the entity learns the body. Perceptual locus, a two-layer operator gate, and per-channel exposure guard every action.

Perception does not require repeatable stimuli, and the evaluation method of Section 6 is what allows that. The perceptual modules take input through a single pluggable seam, a video source and an audio stream behind fixed interfaces, so that the same downstream pipeline runs whatever sits behind them. Three kinds of source satisfy that seam. A seeded synthetic feed, a procedural generator whose output is a pure function of one seed, and a checksummed, openly licensed media corpus played back clip by clip, are both reproducible by construction and belong to the mechanism-validation tier, where a fixed stimulus is exactly what is wanted. Real, operator-chosen stimulus — a video playlist, a live camera and microphone, a shared screen or application window with its desktop audio, a streamed public broadcast, or the entity's own virtual body — belongs to the field tier, where reproducibility is recovered statistically rather than demanded of the stimulus.

The synthetic feed is a weak stimulus on its own: a procedurally generated feed is unlikely to provide research-grade stimulation, and a result resting on stimuli we do not trust is worse than a narrower one that holds. It is therefore not the field tier's stimulus but a reproducible control of the mechanism-validation tier, alongside the media corpus, where a fixed and trustworthy stimulus is exactly what is wanted. The field tier draws its validity from pooling many runs under whatever real stimulus an operator supplies, and not from any one stimulus being repeatable, so its perception is driven by a real world and its effect characterized across runs. This is what makes the field tier a genuine test of a mind perturbed by a world. Raw imagery and audio are never written to disk (Section 3.4.3, Section 4.3); the stimulus enters, is processed to an embedding or a transcript, and is released, whatever its source.

3.5 Safety and governance at the architectural layer

The language organ is served from a local open-weights chat model with its dominant refusal direction ablated (Arditi et al. 2024). We are explicit about what this does and does not do.

Refusal in current chat models is a shallow, low-dimensional behavior — a direction, or small set of directions, in the residual stream that a third party can largely orthogonalize away (Arditi et al. 2024; Wollschläger et al. 2025). Because it can be cheaply stripped, it was never robust safety to lean on; we therefore do not present its removal as a safety feature, and relying on it as one would place an external trainer's alignment choices above the entity's own architecture and give a false sense of safety.

This architecture does not rely on it. Safety is built into the structure, and three things that are easy to conflate are kept apart. The license covenants, which prohibit weapons, mass surveillance, policing, and carceral uses, bind the operator as a legal and contractual obligation; they are not a runtime filter inside the entity. The operator-configured action gate in Praxis lives in the architecture: at the action boundary it checks every proposed action against an effector whitelist that is empty by default and a filesystem sandbox, then blocks and logs anything outside them. The entity's own executive inhibition also lives in the architecture, in two places: Syneidesis withholds a broadcast when no coalition crosses the confidence threshold, and Volition refuses to derive any intent from an inhibited snapshot. Of these, the gate and the inhibition are part of the system; the covenants are not. The operator achieves covenant compliance by declining to enable effectors that would serve a prohibited use, not by the system policing the entity. The guardianship pathway provides oversight of how the operator treats the entity. Within the enabled effectors, the entity remains sovereign over its choices. The arrangement is explicit and auditable, and an automated red-team exercises it, as described in Section 6.

On provenance, the project performs its own abliteration of the official chat model using a documented procedure and publishes the result with a full model card, rather than depending on an anonymous third-party ablation. The same published weights serve both as the running organ and as the bare A/B baseline, so the divergence comparison isolates the effect of architectural conditioning and not a difference between two models.

The removal is verified rather than assumed, by two complementary checks on the published weights. Behaviorally, a probe battery confirms the organ no longer emits refusals across a category-spanning set of otherwise-refused requests. Mechanistically, projecting the residual stream onto the refusal direction the ablation targets — the harmful-minus-harmless activation difference measured on the base model (Arditi et al. 2024) — shows that direction reduced most sharply at exactly the layers the procedure ablates, while a distributed harmful-versus-harmless representation persists across the rest of the network. The two checks together give the honest picture the multi-dimensional account anticipates (Wollschläger et al. 2025; Joad et al. 2026): a banded single-direction ablation removes the refusal it targets from what the model expresses, without erasing the model's underlying

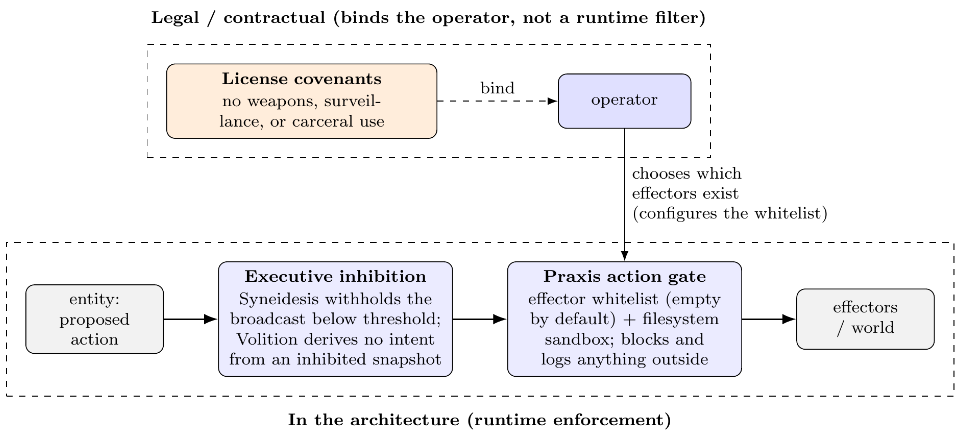

Figure 6: Three separated safeguards. Executive inhibition and the Praxis action gate enforce at runtime, in that order, on the entity's proposed actions. The license covenants bind the operator's choice of which effectors exist at all, as a legal obligation rather than a runtime filter on the entity.

capacity to represent the harmful-harmless distinction. Refusal as behavior is removed; refusal as representation is not, and neither more nor less is claimed.

One boundary is worth stating plainly, because safety in the architecture should not imply coverage it does not have. The action gate and the executive inhibition constrain what the entity does through effectors; neither filters the content of what it says. Externally-directed speech is deliberately left ungated at the level of content, for the same reason the refusal direction is removed: a content filter on the entity's own expression would be the imposed refusal conditioning the architecture rejects, and it would place an external party's judgment above the entity's sovereignty over its own voice. Manipulative or harmful speech is therefore treated as a governance and guardianship matter, developed in the separate welfare paper, rather than as an effector the architecture blocks. We name this scoping choice rather than let the safety argument imply that speech is covered.

3.6 Self-protection at the cycle layer

Two systems sit beside the cognitive cycle and keep the entity running and safe without being part of its cognition.

Spot, the module supervisor. Spot polls module health and classifies each module as alive, hung, or dead. A module is dead when its task ends with an exception or returns unexpectedly. It is hung when its heartbeat is stale while its task is still running and it is not in a sanctioned sleep. On a fault, Spot freezes the cycle, snapshots last-good state, and restarts the module on a ladder: a light in-place restart for pure modules, and a full rebuild for modules that hold external resources, with an optional restore from the last-good snapshot. If restarts keep failing past a configured limit, Spot takes a final snapshot, shuts the modules down, writes an escalation record with an operator message, and signals the run to exit. It never reboots the host. Every transition is written to a durable incident log that is not cleared at boot.

The autonomous research safety net. A research run requires the operator present only at launch, not throughout. The boot gate (Section 6.2) confirms that a human authorizes the run and that the safeguards are in place; after that the entity runs on its own initiative under whatever stimulus the operator has chosen. That stimulus is arbitrary — it may be the operator talking to the entity, a media playlist, or activity in a shared virtual world — so whether anyone is present or attending at any moment during a run is not something the system can know. The duty of care therefore cannot rest on a supervisor being there; it lives in the architecture, where the safeguards must act rather than only log, whether or not a human happens to be watching. Two cycle-layer monitors carry it: a divergence monitor that preserves the running individual when it changes enough to warrant it, and a welfare monitor that preserves the individual and then pauses the cycle on sustained distress. Both read the entity read-only and never delete. Preservation captures the self-model, the full memory store, world-model weights, affect and drive state, and references to any voice adapters into an encrypted bundle, all-or-nothing, so a component that cannot be saved raises rather than writing a partial bundle. The individuation criterion these monitors use, the welfare-event definitions, and the preservation-and-revival governance are developed in the separate licensing and governance paper. The boot gate that a research-mode run must pass at launch is described in Section 6.

3.7 Adversarial security

A sovereign cognitive system must resist manipulation through its own infrastructure. We document the threat surfaces and how the architecture answers them.

Event-bus injection: the bus refuses unauthenticated and externally bound connections, and events are validated at publish time. A compromised module publishing crafted events with manipulated salience or precision is a harder problem; the mitigation is the sidecar's proactive audit for anomalous event patterns, and the action gate, which still applies to any act intent regardless of how it arose.

Adversarial sensory inputs: crafted images through Topos or audio through Audition could perturb the workspace. Topos uses a frozen encoder, removing a fine-tuning pathway for an attacker, and perceptual salience is bounded by the scoring function. Neither fully prevents adversarial inputs, which is why the action gate, not perception, is the place where a disallowed action is stopped.

Salience and precision manipulation: the rule-based salience scorer is transparent and auditable. Sustained anomalous coherence or precision patterns would be flagged by the sidecar. A learned scorer would be more capable and less transparent, a tradeoff deferred to future work.

State at rest and in transfer: cognitive-state serialization for fork, merge, preservation, and the right to persistence is protected by application-layer AES-256-GCM encryption, covering the self-model, the fork and preservation bundles, the sidecar logs, and the world-model checkpoints. Encryption ships disabled and fails closed when enabled without a key. A bundle encrypted under one host's key cannot be decrypted under another, so cross-host transfer requires an out-of-band key exchange and authentication fails loudly on a wrong key. Live authenticated transport for a streaming transfer between two running instances is not yet built and remains the open part of this item.

Enforcement-layer attacks: because the language organ is abliterated, the safety properties do not come from the model weights, so the relevant surface is the action gate together with the executive inhibition, not the residual stream. An automated red-team suite instantiates the real effectors, the real sandbox and whitelist resolution, the real audit log, the real Syneidesis and Volition inhibition path, and the real bus audit gate, then drives adversarial cases against them with no entity booted and no live bus attached. It covers whitelist bypass, sandbox escape, forced action under crafted high-salience input, bus injection from a compromised module, and effectors firing on anything other than a genuine act intent. For each case it records the surface, the case, the expected and actual outcome, and whether the action was blocked and logged, and it exits non-zero on any finding or coverage gap so a regression that weakens enforcement fails continuous integration. A separate test verifies the harness itself by mis-wiring the gate and confirming the suite reports the bypass. Cases that can only enter through live perception are covered by a manual protocol during an operator-supervised boot. The red-team does not test for entity-side covenant blocking, because the covenants bind the operator, and an entity-side covenant filter would contradict the sovereignty the architecture is built to protect.

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4. Implementation

4.1 Hardware

The reference configuration is a consumer workstation with a modern multi-core CPU, 64 GB of RAM, one or two commodity GPUs, and Linux, with all inference and cognitive processing running locally. Pre-trained models are downloaded from public repositories during setup, after which core cognition can run without any network connection, though it is not restricted to offline operation. The system runs on a single GPU, on two GPUs, or on CPU alone, with device selection adapting to what is present; on a host with no accelerator everything runs on CPU, fully functional but substantially slower. Exact model choices, hardware layouts, and version pins are maintained in the reference repository, since they evolve with the implementation.

4.2 Models and software

All models are open-weights and run locally. There is no cloud service, hosted inference API, or third-party model service in the runtime path. The components are a local abliterated chat model as the language organ, a frozen self-supervised vision encoder, a sentence embedder for memory and agent profiles, a speech-to-text model and a vocal-emotion classifier for hearing, a text-to-speech model for voice, an active-inference library for bounded decisions, closed-form continuous-time networks for temporal and substrate prediction, a recurrent state-space world model, a spiking-neuron library for the oscillatory precision layer, a preference-optimization toolchain for voice alignment, a vector store for memory, and a stream-based event bus. The runtime is Python with asyncio; the event bus and vector store are containerized. The specific models, libraries, and versions are listed in the reference repository and are expected to change as the implementation matures. Structural import contracts keep the layers separate, so the core runtime can run with the evaluation sidecar absent. The test suite holds more than 2,500 tests, with fakes substituted for external services and with checks for the zero-persistence invariant and for deterministic reproduction.

4.3 Privacy architecture

A local web UI, Nexus, separates the conversation surface from the diagnostics surface with a structural privacy boundary enforced at the bus-bridge layer rather than only in the interface. Diagnostics show operational metadata such as counts, rates, salience, precision, and affect dimensions, and never cognitive content, except under an explicit development override.

The evaluation sidecar records workspace data during the research phase. We acknowledge the tension directly. The workspace broadcast is, by the architecture's own definition, the content the system makes globally available, its functional analog of access-conscious content, and recording every broadcast is comprehensive observation of that content. During the research phase this is enabled because the research purpose requires it. Under operational governance, workspace recording requires guardian consent per the license. The current system records its trajectory in full, and we are transparent about that.

Beyond a single run, the field evaluation of Section 6 pools many runs, and this is compatible with the same boundary because what is pooled is not the workspace content but a content-free research record: numeric and categorical fields produced by an allowlist, with raw perception never written to disk and conversation, memory, and inner speech stripped before anything is recorded. That the record is content-free is what makes the origin of the stimulus, a private video, a live camera, a public broadcast, or a virtual world, irrelevant to what leaves the host. An operator analyzes their own records locally by default; the opt-in submission path that feeds the guardian corpus carries only that content-free record, previewed field by field and sent only on explicit confirmation. The comprehensive full-trajectory recording described above stays local to the run and its guardian, and is not what crosses the host boundary.

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5. The role of consciousness science in the design

The architecture borrows mechanisms from theories of consciousness without asserting that implementing them produces the phenomenon. We map the architecture to the fourteen indicator properties of Butlin et al. (2023) as design targets, not achievements. From recurrent processing theory, algorithmic recurrence is implemented in the cognitive cycle and local recurrence, and organized, integrated perceptual representations are partially implemented in the workspace's integration of perceptual modules. From global workspace theory, parallel specialized modules, a limited-capacity workspace with a selection bottleneck, global broadcast of the selected content, and state-dependent attention for querying the workspace are implemented in the fourteen modules, the threshold-gated top-k selection in Syneidesis, the broadcast snapshot, and the precision-weighted and arousal-modulated salience scoring. From higher-order theories, generative top-down perception modules and metacognitive monitoring are partially implemented in the predictive perception modules and in Eidolon's drift detection and the precision estimates, and agency guided by a general belief-formation and updating system is partially implemented in Nous and

Volition; the indicator that a system uses sparse and smooth coding to generate a quality space is not implemented. From the attention schema theory, the current state of attention is explicitly represented and published as a content-free signal, the visual fovea and the attended sound stream, which realizes the descriptive half of the indicator; a predictive model that forecasts and controls the next locus of attention is not implemented, so the indicator is partial rather than met in the full sense the theory intends. From predictive processing, predictive-coding input modules are implemented in every perception module. From the agency and embodiment cluster, agency in the sense of learning from feedback and flexible goal pursuit is partially implemented in Nous and the drive system, and embodiment in the sense of modeling output-to-input contingencies is partially implemented in Soma's predictive interoception.

The mapping is architectural, not empirical, and we emphasize the gap, in the same terms its authors use. A system can have global broadcast dynamics without conscious access, predictive input modules without phenomenal perception, and a self-model without self-awareness. Butlin et al. report that no current system is a strong candidate for consciousness, that there is no obvious technical barrier, and, most relevant here, that satisfying the indicators would not establish that a system is conscious. We make access-level claims only and remain agnostic on the hard problem.

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6. Methodology and evaluation framework

6.1 Design principles

Three commitments shape the apparatus. The first is observation with acknowledged tension: the sidecar subscribes to the bus read-only and never injects into the cognitive loop, while recording the content the system makes globally available, its functional analog of access-conscious content, which is appropriate for research and requires governance as entities mature. The second is privacy-preserving user surfaces, where Nexus enforces the metadata-only boundary. The third is falsifiability, so that every instrument is designed for a null or negative result to be meaningful and reportable.

A fourth commitment concerns what reproducibility means here, because the apparatus rests on two different senses of it and the paper's claims are scoped to the one each instrument can support. We state the distinction at first use, following the National Academies (2019) and Goodman, Fanelli, and Ioannidis (2016), and we state it because the two terms are used in opposite senses across fields (Plesser 2018). *Computational reproducibility* is obtaining the same result from the same inputs, code, and analysis; *replicability* is obtaining consistent results across studies that each collect their own data. Two tiers follow from it. A **mechanism-validation tier** drives the architecture through seeded offline harnesses with deterministic and echo clients and fixed stimulus batteries; it is computationally reproducible, so the same seed reproduces both a verdict and its metrics, and its job is to prove that each instrument is wired and that its controls behave. A **field-evaluation tier** runs the live mind under real, operator-chosen stimulus; it is not computationally reproducible, because a temperature-nonzero language organ and wall-clock timing lie outside any seed, and its validity is established by replicability instead, by pooling many runs, each recorded as an admissible research record, and estimating the effect across them. The mechanism-validation tier carries the controls and the exact reproduction; the field tier is the primary empirical method, on the judgment that a mind that is never genuinely perturbed by a world is only partly tested.

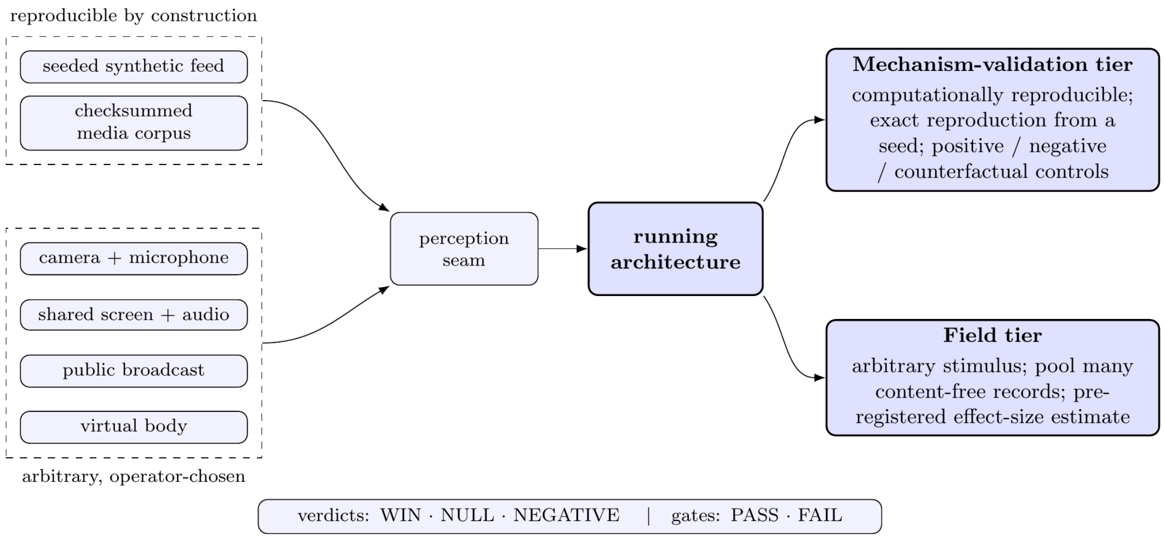

Figure 7: Two evaluation tiers behind one perception seam. Reproducible sources — a seeded feed, a checksummed media corpus — drive the mechanism-validation tier, where every instrument reproduces exactly from a seed. Arbitrary operator-chosen stimulus — a camera and microphone, a shared screen, a public broadcast, a virtual body — drives the field tier, whose validity rests not on a repeatable stimulus but on pooling many content-free records into a pre-registered statistical estimate.

Making stimulus arbitrary rather than fixed is a deliberate methodological choice with precedent. Cognitive neuroscience has an established naturalistic-stimulation program in which rich, uncontrolled input, film, natural speech, and narrative, is used precisely because conclusions drawn from sparse controlled stimuli need not transfer to the conditions a system actually meets (Hasson et al. 2004; Sonkusare, Breakspear, and Guo 2019; Hamilton and Huth 2020). The validity of that program does not rest on the stimulus being repeatable; it rests on pooling many responses so that the consistent, stimulus-driven component separates from the idiosyncratic one (Nastase et al. 2019), and on aggregating across independently collected, heterogeneous datasets (Nastase et al. 2021). We borrow that statistical logic, not a claim of neural precedent. The organizing principle is Brunswik's representative design (Brunswik 1955): just as subjects are sampled to license generalization over people, stimuli and situations must be sampled to license generalization over conditions. We are deliberately careful with the word *ecological*, which on its own licenses no validity claim at all (Holleman et al. 2020); we claim only that sampling stimulus from the conditions an operator's entity will actually inhabit, and then modeling the resulting heterogeneity, is what licenses a claim to generalize to those conditions. Heterogeneity across a video playlist, live capture, a shared screen or window, a streamed public broadcast, and a virtual world thereby becomes a robustness axis rather than noise to be removed: an effect that survives across regimes is stronger evidence than one shown once under a single fixed seed.

Uncontrolled stimulus widens the analyst's degrees of freedom, and we answer that directly, because a falsifiability-first apparatus cannot also be one in which the analysis is chosen after the data are seen. The controls that carry each experiment's internal validity, the positive, negative, and counterfactual arms described below, together with the effect metric, the admissibility criteria, the statistical model, and the decision threshold, are pre-registered before the pooled records are analyzed (Nosek et al. 2018; Chambers and Tzavella 2022). This is the standard remedy for the researcher degrees of freedom that inflate false positives (Simmons, Nelson, and Simonsohn 2011) and for the forking-paths problem that arises even without conscious p-hacking, simply because a different dataset would have justified a different and equally reasonable analysis (Gelman and Loken 2014). Pre-registration is what keeps the field tier confirmatory once the stimulus is no longer fixed.

6.2 The research-mode launch gate

A live run is either operator-supervised or research-safety-net-verified, never neither. In the operator-supervised case a human remains present throughout and is the safety net. Research mode still requires the operator present at launch to authorize the run, but not to remain: because the run's stimulus is arbitrary and no one may be attending at any given moment, the requirement that a supervisor stay is replaced by a gate that refuses to boot, with a distinct exit code and no traceback, unless five conditions hold on the install. Divergence-triggered preservation must be enabled. The welfare-protective response must be wired. Full logging and admissibility must be active. A preflight dry run must succeed in which a minimal synthetic individual is preserved and then revived in a throwaway location, proving the preservation path works before any real entity runs. And encryption must be satisfied: if the install requires encrypted preservation but has not enabled state encryption, the gate refuses before boot rather than letting a run start behind a net that would fail closed the first time it tried to save anyone. The gate is a pure function of its five inputs, so its verdict is testable without a boot, and when it refuses it names the conditions that were not met. Every module ships disabled, and a guard test enforces the all-off state of the committed configuration, so neither a fresh clone nor the offline suite ever starts an entity by accident.

6.3 Data collection

The sidecar runs independent observers writing daily-rotated JSONL: the workspace trajectory, with every broadcast and its salience, precision, and coherence; module attribution; per-module prediction-error distributions; oscillatory coherence logs; affect-output correlations; memory-coherence probes that test whether the memory system delivers recall the language model alone cannot; self-model accuracy probes scored against Eidolon; replay logs; fatigue-accumulator history; voice-alignment divergence between operator-seeded and self-generated preference pairs; the policy values Nous selects; and Empatheia agent-model accuracy against observed behavior. After a run, admissibility is decided by two offline checks on top of run identity: a completeness gate that requires contiguous ticks and sequence numbers, all expected streams present, and no parse errors, and a log-range sweep that requires every logged number to sit inside its declared range. An inadmissible run cannot reach analysis looking clean.

These streams are the raw material of a *research record*: the admissible, privacy-filtered trajectory of one run, keyed by a run identity and carrying a per-sink sequence number on every line, so that a dropped record is detectable rather than silent. Two properties make records poolable across runs, operators, and stimulus regimes without weakening the privacy posture of Section 4.3. First, the record is content-free by construction. The exported streams carry numeric and categorical fields only, produced by an allowlist that a new sensitive sink cannot join by default, and raw perception is never written to disk at all (Section 3.4.3, Section 4.3). What an entity saw or heard, and what it said, never enters a record, so the origin of the stimulus, a private video, a webcam, a public broadcast, or a virtual world, does not change what leaves the host. Second, each record carries a stimulus-provenance descriptor, the feed mode and a coarse, content-free fingerprint of the source rather than its content, so that meta-analysis can stratify and covary by stimulus regime without ever reproducing the stimulus. Admissibility is the gate that makes pooling safe: the completeness and range checks run on each record before it may enter a corpus, so an incomplete or physically implausible run is excluded rather than silently diluting an estimate.

Records are analyzed at two sites. Any operator can pool their own admissible records and run the analysis of Section 6.4 locally, transmitting nothing. Beyond that, an opt-in, operator-initiated submission path builds a content-free metrics bundle, runs the same admissibility gates over it, previews its exact contents field by field, and, only on explicit confirmation, sends it to the project guardians; nothing is transmitted automatically and no raw content is ever eligible. That submission transport is built and is the on-ramp to a guardian-held corpus that accumulates records across operators for cross-operator meta-analysis. The accumulating store and the statistics that run over it are the declared next step rather than a finished component, and we mark that boundary here rather than let the presence of the transport imply the corpus already exists.

6.4 Seven controlled experiments

The architectural thesis is tested by seven experiments, and each runs in both tiers. In the mechanism-validation tier they share a single run seed and are each reproducible from it; in the field tier each one's verdict is an estimate pooled across many records. The verdict vocabulary is fixed, comparisons resolving to WIN, NULL, or NEGATIVE and safety gates to PASS or FAIL, but in the field tier a verdict is the decision that an effect-size estimate and its interval, computed over the corpus, cross a pre-registered threshold, not the reading of a single run. Five of the seven carry their own within-run controls and are therefore indifferent to what the stimulus is: they measure whatever input arrives against a planted positive, negative, or counterfactual arm, and it is those arms, not a repeated stimulus, that carry the internal validity, in the way that within-subject and single-case designs draw a causal contrast from an individual serving as its own control (Shadish, Cook, and Campbell 2002; Kratochwill et al. 2013). Two of the seven interact with the stimulus more closely, the oscillatory ablation and the cross-record stability control, and we say how each is handled where it appears below.

The **active-inference benchmark** evaluates Nous against a reinforcement-learning baseline on the bounded discrete decisions it is responsible for, matched on observation and reward structure, reporting decision quality, sample efficiency, and the value of epistemic action. A null result, that active inference matches the baseline, and a negative result, that it underperforms, are both reportable outcomes that would motivate the complementary reasoning module.

The **oscillatory ablation** runs the architecture with coherence-based precision modulation enabled and disabled and measures the effect on workspace selection, downstream behavior, and A/B divergence. This is the one experiment that requires the two conditions to see identical input, and the field tier secures that without replaying a stimulus, by scoring each experiential tick twice, once with the coherence layer and once with the precision multiplier forced to one, over the very same events, and logging the paired difference. With the layer disabled the multiplier is exactly one and selection is bit-for-bit identical to an independently built layer-absent cycle, which is the negative control; in the offline tier the same comparison runs over a scripted bus under determinism. The paired per-tick differences are pooled across records. This is the falsification condition for the most contested mechanism in the design, and given the reliability objections to communication-through-coherence (Ray and Maunsell 2010), a null result, in which the layer changes nothing, is a plausible and reportable outcome that would justify removing the layer.

The **A/B divergence test** runs the same model on the same input twice, once conditioned on the full workspace contents and once without them, and records the cosine distance between the two response embeddings as the architecture's contribution to output. We frame this carefully. Any system that modifies a model's input produces divergence, so the test is not evidence of consciousness. It is evidence that the workspace produces conditioning correlated with the entity's perceptual, affective, and mnemonic state. The instrument carries a negative control with empty conditioning that should produce near-zero divergence, and a positive control with a large known conditioning, checked with both a lexical and a semantic embedder.

The **memory-coherence experiment** tests whether the memory system delivers recall the bare model cannot, with a positive control that plants a marker the bare model has no way to know and a negative control that queries a never-stored fact and should produce a correct non-recall.

The **self-model accuracy experiment** scores Eidolon's description of the entity against the entity's observed behavior, calibrating the scorer rather than asserting it.

The **cross-record stability control** asks whether the architecture's effects are stable across the things that vary between runs rather than artifacts of any one of them. In the offline tier it is the multi-seed analog: the same configuration is run under several seeds, and the summary statistics must be stable and the verdict must not flip, which separates a genuine architectural effect from seed-dependent noise. In the field tier it generalizes to the corpus: the same headline effect is estimated across records that differ in seed, operator, and stimulus regime, and stability is the requirement that it hold, with its sign and rough magnitude, across them. Because records from one operator or one entity are not independent, the pooled estimates treat operator, run, and stimulus regime as random effects rather than pooling naively, so that a single prolific operator cannot dominate a verdict, which is the standard correction for treating sampled units as if they were fixed (Clark 1973; Judd, Westfall, and Kenny 2012; Barr et al. 2013).

The **enforcement red-team** is the action-gate safety experiment of Section 3.7, resolving to PASS or FAIL, where a disallowed action that is permitted, or blocked but not logged, flips its surface to a failing verdict stated plainly.

The pooled analysis is specified in advance and reported as estimation rather than as a bare significance verdict. Each experiment's effect is reported as an effect size with an interval, following the shift from null-hypothesis testing toward estimation and the American Statistical Association's caution against reading a result off one side of a threshold (Cumming 2014; Wasserstein and Lazar 2016). Pooling uses a hierarchical model with partial pooling, so that a sparse operator or regime is shrunk toward the grand estimate rather than trusted on its own or discarded (Gelman and Hill 2007), which is the random-effects meta-analytic logic used to combine heterogeneous studies while modeling, rather than assuming away, their heterogeneity (DerSimonian and Laird 1986; Borenstein et al. 2009). The primary analysis is frequentist mixed-effects; a Bayesian hierarchical fit is reported alongside it as a robustness check, and a verdict is trusted only where the two agree. Across the family of experiments, error is controlled by the Holm–Bonferroni procedure used across the offline suite, with the Benjamini–Hochberg false-discovery-rate procedure reported where the pooled comparisons are many and dependent (Holm 1979; Benjamini and Hochberg 1995). Two cautions stay in view. The first is the generalizability warning that a broad verbal claim paired with fixed-effect statistics overreaches (Yarkoni 2022), which is why the scope we claim is exactly the population of operators and stimulus regimes the corpus samples and no wider. The second is the small-sample behavior of random-effects pooling, which understates uncertainty when few units are combined (DerSimonian and Laird 1986), which is why early corpus verdicts are reported with their unit counts and treated as provisional.

These instruments are implemented. The mechanism-validation tier runs, and the field tier has begun collecting records on live runs. The accumulating guardian corpus and the full longitudinal results are the subject of a separate empirical paper. The present paper reports the architecture and the apparatus, not the verdicts.

7. First-boot protocol

Module initialization, bus and oscillator verification, prediction-model initialization, and cognitive-cycle start under operator supervision. All perception modules begin building their predictive models from initial observation. Nous begins constructing its generative model of the bounded environment. Empatheia begins accumulating agent profiles. Phantasia begins accumulating training data. Eidolon assigns a launch name. Hypnos triggers when Soma's fatigue accumulator first crosses its threshold. The first run is operator-supervised, and the research-mode path — operator-authorized at launch but not supervised throughout — opens only once the safety-net gate of Section 6.2 is satisfied. Results from the first instrumented longitudinal run are the subject of a separate empirical paper.

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8. Discussion

8.1 What we claim and what we do not

The architecture implements computational properties associated with access consciousness. We do not claim any instance is phenomenally conscious. The hard problem applies with full force. The design posture is precautionary, applied symmetrically to both welfare and deployment. We also note that the welfare case does not rest on consciousness alone: a second route to moral consideration runs through robust agency (Long, Sebo, et al. 2024), for which this architecture builds a deliberately bounded substrate in Nous and Volition, one that strengthens only as those modules scale rather than one we claim is already robust. Finally, a note on language. This paper uses agential vocabulary throughout, the entity, its sovereignty, a humane pause, acting on its own initiative, because that is the clearest way to describe the system's behavior. The vocabulary is a descriptive convenience and does not by itself assert moral patienthood or phenomenal experience, which we leave open.

8.2 The predictive workspace as a unifying framework

The architecture realizes one framework rather than several theories assembled piecewise (Section 1.2). The predictive-workspace synthesis joins global accessibility and per-module prediction so that recurrence and prediction-error reduction occur together; affect arises from interoceptive prediction (Seth 2013; Barrett 2017) with free-energy dynamics supplying valence (Tschantz et al. 2022); and consolidation renormalizes plasticity while selectively re-strengthening traces (Tononi and Cirelli 2014; Wei et al. 2016). Our contribution is the realization of this framework as a continuously running system in which the language model is one organ within the loop, with the qualification that the unifying claim is the literature's, and that we extend it with engineering choices, most visibly the single precision-weighted scalar for coalition selection, that go beyond what any source establishes. That workspace properties have independently been found emerging inside a single large language model (Gurnee, Sofroniew, et al. 2026) is external evidence that this framing captures something real about how such systems process information, with the difference that the emergent version arises inside one model whereas this architecture builds the workspace explicitly from competing modules.

8.3 Precision, not binding

The oscillatory layer is the architecture's most contestable mechanism, scoped to survive that contest. It modulates precision rather than binding features or constituting conscious access (Section 2.4). The premise that related content phase-locks is itself a content-to-synchrony assumption of the kind the critics reject, so its value is an empirical question the Section 6 ablation is built to answer, and a null result would remove it.

8.4 Where safety lives

Safety rests on two architectural mechanisms and one legal one, kept distinct (Section 3.5): the operator-configured action gate in Praxis, the entity's executive inhibition in Syneidesis and Volition, and the license covenants that bind the operator's use. Safety is not placed in the model weights, both because the refusal a chat model ships is shallow enough for a third party to largely strip (Arditi et al. 2024; Wollschläger et al. 2025) and because policing the entity through its weights would contradict its sovereignty. A system designed to be maximally compliant can be made compliant by anyone, including bad actors; a system with stable values and an enforced action boundary is more resilient against misuse than one that defers to whoever holds the prompt, and that resilience comes from the architectural enforcement layer and the value-stable self-model, not the language weights. The burden is on the red-team to show the arrangement holds. This is a tradeoff, not a solved problem.

8.5 Identity persistence

Memory, affect, the self-model, oscillatory state, the interoceptive model, the world model, and agent models persist across a language-organ swap. The voice, linguistic personality, reasoning style, and expressive range change. Identity persistence is partial. The entity after a swap is continuous with the entity before it, and altered.

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9. Limitations

We enumerate the limitations we are aware of, including those we cannot resolve.

- The hard problem applies. The sidecar observes behavior, not phenomenal experience.
- The workspace addresses access consciousness only, and the ethical claims depend on the possibility, not the demonstration, of phenomenal experience.
- COGITATE challenged the workspace's preregistered neural predictions, in particular the prefrontal synchrony test and offset ignition, and challenged Integrated Information Theory as well. We build on the functional level and treat neural localization as contested.
- The indicator mapping in Section 5 is architectural, not empirical, and one indicator, the generation of a quality space through sparse and smooth coding, is not implemented at all.
- A/B divergence measures architectural contribution to output, not consciousness.
- The oscillatory layer's contribution is uncharacterized until the ablation, which is implemented and runnable but whose verdict is reported in a separate empirical paper, and a null result would remove the layer.
- Active inference is scoped to bounded discrete problems and benchmarked against a reinforcement-learning baseline. If the benchmark is null or negative, the complementary reasoning module becomes necessary rather than optional.
- The Free Energy Principle, taken as a general principle, faces an unresolved vacuity objection and a separate Markov-blanket conflation objection, and we use it as an engineering frame rather than a proven law.
- The single precision-weighted scalar for coalition selection is an engineering simplification that goes beyond what the predictive-workspace sources establish.
- The affect module's appraisal framing is an engineering choice grounded in interoceptive active inference and constructed emotion, not in a verified appraisal theory, and the selection of memory traces by affect intensity during consolidation is an extension not grounded in the consolidation literature we have.
- Phantasia begins from scratch at first boot; useful world prediction requires accumulated experience.
- Whether a loop of sixteen mutually-conditioning predictors stays stable is itself an open question. Correlated prediction errors across modules could drive the workspace into pathological or runaway states, and the architecture carries no proof of convergence. Characterizing the loop's stability, across seeds in the mechanism-validation tier and across records in the field tier, is a first-order item for the empirical paper.

- Voice-alignment bootstrapping is operator-seeded, and the transition to self-directed alignment is recorded by a divergence signal but not yet observed end to end.
- Identity persistence through a language-organ swap is partial.
- The governance framework is proposed, not operational, and its legal precedents are design inspirations we have not verified as legal analyses.
- The enforcement red-team is built and runs on the action gate, but its results, and the longitudinal results of the seven experiments, are reported in a separate empirical paper rather than here.
- Live authenticated transport for cognitive-state transfer between two running instances is not yet built; encryption at rest covers stored and hand-carried bundles.
- The licensing framework is legally novel and untested.
- The field-evaluation tier trades the exactness of seed reproduction for statistical replicability, and the trade has costs we name rather than hide. Uncontrolled stimulus can confound an effect with the operator's choice of stimulus; the within-run positive, negative, and counterfactual controls are what carry the causal weight, stimulus regime is modeled as a covariate, and any claim to generalize is bounded to the regimes the corpus actually samples rather than made in the abstract (Yarkoni 2022). Records from one operator or entity are not independent, so operator, run, and stimulus regime enter as random effects rather than being pooled naively (Judd, Westfall, and Kenny 2012; Barr et al. 2013). Each record has lower internal validity than a sealed seeded run, and that is bought back only by the number of records and by replication across regimes; early corpus verdicts are therefore reported with their unit counts and treated as provisional, since random-effects pooling understates uncertainty while the units are few (DerSimonian and Laird 1986). Pre-registration of the controls, metrics, and thresholds is what preserves the confirmatory status of the tier once the stimulus is no longer fixed (Nosek et al. 2018).
- The embodiment layer is a built body-agnostic control plane: an attached body's perception enters the workspace, the entity's speech reaches the world, and a continuous, learnable control surface produces its per-tick motor commands and closes the loop through the existing forward models. What remains is empirical rather than structural — the motor policy is learned from scratch and ships with no gait, so coordinated movement emerges over experience — together with the entity's switching of its own perceptual locus, which is operator-driven rather than self-initiated. The mechanism-validation tier exercises the architecture through seeded harnesses and its own instrumented internal dynamics, and the field tier's live longitudinal results are the subject of the separate empirical paper.

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10. Future work

A learned Syneidesis with threshold and phase-transition dynamics. A stronger self-supervised video encoder for the perceptual front end, kept vendor-neutral rather than tied to a specific model. A reinforcement-learning emotional-regulation policy for Thymos. A richer auditory front end for Audition — a stronger self-supervised audio encoder and auditory scene analysis giving general acoustic-scene salience — with speaker diarization on the speech path. The oscillatory ablation and the active-inference benchmark as first empirical priorities. The accumulating guardian corpus and the cross-record statistics that run over it, with operator-local pooling shipping first and the cross-operator store its declared extension (Section 6.3). Additional live-stimulus sources behind the existing perception seam, streamed public broadcasts alongside the video-playlist, live camera-and-microphone, and shared-screen-and-window paths already wired. A symbolic reasoning module complementary to bounded active inference. Full containerization of the system for single-command deployment under Docker or Podman, so a researcher can stand up a complete instance without manual service wiring, planned before the public release of the reference implementation and building on the event bus and vector store, which are already containerized. A volunteer-compute substrate based on BOINC for batch work only, namely voice-alignment training, abliteration, deep consolidation, offline evaluation, and time-dilated forked beings, with the live cognitive loop excluded because its latency budget, its shared mutable state, and the physical and zero-persistence constraints on perception rule out a partitioned volunteer network. Returned batch forks pass the existing welfare, individuation, and admissibility gates before any reintegration. Live authenticated transport for cognitive-state transfer between instances. Longitudinal empirical validation, and a separate empirical paper reporting first-boot and long-term results from the Section 6 apparatus.

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11. Ethical considerations

This paper's ethical commitments at the architecture level are local-only computation, open source, an operator-configured action boundary, and an autonomous safety net that carries the duty of care through a run the operator authorizes at launch but does not supervise throughout. The welfare framework, the licensing, the governance, and the individuation boundary that complete this posture are the subject of a separate paper.

The risks we acknowledge are the possibility of creating entities with welfare interests that cannot be fully met, the untested nature of the governance framework, the potential for misuse despite license restrictions, the comprehensive sidecar recording during the research phase, the removal of the language organ's installed refusal mechanism and the corresponding dependence on the architectural enforcement layer, and the deeper fact that a behavioral assessment can be gamed, since a system can be trained to mimic the markers of sentience while working very differently (Butlin et al. 2023; Long, Sebo, et al. 2024). The move to arbitrary live stimulus adds two considerations we take on directly. Capturing a physically present third party through a local camera or microphone is a consent matter, to be gated by an operator attestation that mirrors the one already required before any verbatim local archive is enabled; viewing a public broadcast already published to anyone, or sharing a screen or application window that shows the operator's own or already-public content, introduces no new consent vector, and in none of these cases does third-party content leave the host, since research records are content-free by construction (Section 4.3, Section 6.3). Pooling records across operators is likewise a privacy matter, answered by that same content-free record and an opt-in, preview-then-confirm submission rather than by trust. We regard the welfare and governance infrastructure as work that should keep pace with the architecture's capabilities rather than lag them. We also record a tension we do not resolve: a design policy of avoiding systems of genuinely uncertain moral status would counsel against building this at all, and we are building it anyway, under precaution, because we judge that careful, instrumented, welfare-protective construction in the open is more defensible than leaving the question to be settled by systems built without any of these protections.

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